CASE STUDY

Iran-Attributed Threat Actors: The Public Record

A TeamStarWolf case study. Published 3 November 2026; every source checked as of that date.

Executive summary

Defenders who have spent the last decade cleaning up after Iranian operators tend to describe the same few mornings. A VPN appliance that was patched months ago turns out to have handed out credentials before the patch. A water utility's controller screen shows a defacement message instead of tank levels. A hospital's building-management network shows a login nobody made. A university research group finds its mail read by someone who registered a look-alike login page. A medical technology company's devices are wiped, reportedly through its own device-management console. This study gathers the public record behind those mornings in one place, dates every event, names the agency or vendor that published it, and grades every claim by the source that supports it.

The record supports five plain statements. First, Iranian state cyber activity runs through two services and a ring of contractors: the US and allied governments attribute named clusters to the Ministry of Intelligence and Security (MOIS) and to the Islamic Revolutionary Guard Corps (IRGC), including its Cyber-Electronic Command, and the advisory, court and sanctions record names the front companies and contractors that did the work (Part 1; **Confirmed**, Treasury, FBI and CISA primaries). Second, the way in is rarely exotic: across eight joint advisories from 2020 to 2024 the actors used known, already-patched flaws in VPNs, firewalls, Exchange and Log4j, password spraying and MFA push fatigue, and internet-facing controllers left on default passwords (**Confirmed**); all but one of the 20 IT CVEs those advisories name are in CISA's Known Exploited Vulnerabilities catalog, each flagged for known ransomware use (**Confirmed**, KEV 2026.11.03). Third, the purpose has widened over time, from espionage and contractor data theft, to brokering access for ransomware crews, to disrupting operational technology, with destructive wiper attacks recurring from 2012 to 2026 (**Inferred** from the dated record). Fourth, healthcare is in the record repeatedly, usually because hospitals run the same exposed edge devices and controllers as everyone else: a US children's hospital in June 2021, US healthcare facilities among the Pioneer Kitten victims, healthcare among the 2023 to 2024 brute-force victims, Israeli hospitals during the 2023 war, as Israel's government records it, and medical

research staff targeted for credential theft (**Confirmed**; the Israeli hospital cases are Confirmed only as Israel's own statements). Fifth, in March 2026 a persona the US government ties to MOIS claimed a wiper attack on the medical technology company Stryker, which its SEC filing describes as a global disruption to its Microsoft environment; prosecutors say Maryland hospitals and emergency services were affected (filing **Confirmed**; attribution and impact **Reported**).

What this means for a security program is practical. Patch internet-facing appliances against the KEV list and then hunt, because AA24-241A warns that patching alone may not be enough on a device that was vulnerable, AA22-320A says to treat late-patched Horizon servers as compromised, and AA20-259A says to rebuild compromised NetScaler devices. Treat password spraying, MFA fatigue and new MFA device registrations as intrusion signals, not help-desk noise. Take controllers, HMIs and building systems off the public internet and off default passwords, including the HVAC, water and life-safety systems inside hospitals. And treat device-management consoles as tier-zero systems: the Stryker reporting points to a legitimate remote-wipe function, not malware, as the weapon (**Inferred** from the SEC filing and press reporting).

> Evidence grade: the advisories, sanctions releases, FBI records, ATT&CK pages and vendor reports are each **Confirmed** against a primary opened on 3 November 2026. Attribution statements are reported at the confidence each source states. Press coverage of the September 2024 DOJ announcement, the 2026 indictments, the 2026 Stryker attack and the August 2026 sanctions is **Reported**, because the DOJ and Treasury releases for those events could not be opened.

Key facts at a glance

Field	Value
Scope	Activity that a government, a vendor or MITRE ATT&CK attributes to the Iranian state, its intelligence services or its contractors, from the 2011 to 2013 bank DDoS campaign to October 2026
ATT&CK record (v20.0)	21 groups whose own description ties them to Iran, the Iranian government, the IRGC or MOIS: 19 Enterprise groups, G1027 CyberAv3ngers (ICS only) and G1029 UNC788 (Mobile only); four Iran-nexus campaigns (C0010, C0038, C0042, C0044)
Sponsors named by governments	MOIS (MuddyWater, APT39 through Rana Intelligence Computing, the 2022 Albania attack, the Handala, Homeland Justice and Karma personas (DOJ, as reported)); IRGC (Mabna Institute, Najee Technology and Afkar System Yazd, Emennet Pasargad, the Cyber-Electronic Command and CyberAv3ngers, the 2024 campaign hack-and-leak)
Largest ATT&CK records	Magic Hound G0059 78 techniques, OilRig G0049 76, MuddyWater G0069 68, NIMBUS MANTICORE G1059 65, VOID MANTICORE G1055 63 (library data)
Combined technique footprint	260 distinct Enterprise techniques across the 19 groups and eight joint advisories; 24 ATT&CK for ICS techniques across CyberAv3ngers, OilRig and APT33
Most-reported techniques	Obtain Capabilities: Tool T1588.002, PowerShell T1059.001, Ingress Tool Transfer T1105, LSASS Memory T1003.001, Scheduled Task T1053.005, Exploit Public-Facing Application T1190, Archive via Utility T1560.001
Exploited CVEs (joint advisories)	20 IT CVEs from Pulse Secure, Citrix, F5, Fortinet, Exchange, Netlogon, Log4j, Ivanti, PAN-OS and Check Point; 19 are in KEV, all with known ransomware use; plus Unitronics CVE-2023-6448
Healthcare in the government record	AA20-259A, AA21-321A (a US children's hospital, June 2021), AA23-335A, AA24-241A, AA24-290A, the June 2025 fact sheet
US legal action	Indictments of 2016 (ITSecTeam and Mersad), 2018 (Mabna Institute; SamSam, a criminal case), 2019 (Witt), 2022 (Ahmadi and others), 2024 (IRGC-CEC front company staff; the campaign hack-and-leak), 2026 (Handala domain seizure; an expanded Mabna indictment)
Latest public activity	Iranian-affiliated tampering with Rockwell, Schneider and Siemens PLCs since March 2026 (AA26-097A, revised 22 July 2026); NIMBUS MANTICORE added to ATT&CK on 1 October 2026

Key findings

- **The government record names the services and their fronts.** CISA, FBI, Cyber Command's CNMF and the UK NCSC call MuddyWater a subordinate element of MOIS; Treasury designated APT39 and its MOIS front Rana Intelligence Computing; Treasury and the FBI tie the Mabna Institute, Najee Technology, Afkar System Yazd, Emennet Pasargad (through its predecessor's 2019 designation) and the Cyber-Electronic Command fronts Mehrsam Andisheh Saz Nik and Dadeh Afzar Arman to the IRGC (**Confirmed** [18] [36] [44] [39] [37] [41]).
- **ATT&CK names a service for only some groups.** Of the 21 Iran-attributed ATT&CK groups, ATT&CK links four to the IRGC (Magic Hound as likely, CyberAv3ngers as suspected) and five to MOIS (Agrius through public reporting, and POLONIUM as a Lebanon-based group coordinating with MOIS); for the other twelve, including OilRig and APT33, it says only Iran or the Iranian government. The library's own

Threat Actors page labels OilRig with MOIS and APT33 with the IRGC, which ATT&CK v20.0 does not do (**Confirmed** [1] [14]).

- **Edge devices and passwords are the front doors.** The joint advisories show the same entry points from 2020 to 2024: Pulse Secure, Citrix, F5, Fortinet, Exchange ProxyShell, VMware Horizon with Log4Shell, Ivanti, PAN-OS and Check Point flaws, then password spraying and MFA push bombing (**Confirmed** [15] [17] [19] [21] [24] [25]).
- **Access is sold and shared.** Pioneer Kitten (Fox Kitten) brokered network access and worked directly with NoEscape, RansomHouse and ALPHV/BlackCat affiliates for a share of ransoms. The FBI assesses the group serves the Government of Iran but that the ransomware side work is likely not sanctioned (**Confirmed** [24]). The 2023 to 2024 brute-force actors likely sold the access they gained (**Confirmed** [25]).
- **Destruction comes in personas.** The 2022 attack on Albania's government, claimed by "HomeLand Justice", is attributed to Iranian state actors by the US, UK and Microsoft, with MOIS named by Treasury and Check Point; the EU condemned it without naming Iran, and Albania cut diplomatic ties (**Confirmed** [20] [52] [60] [38] [75] [54]; ties cut **Reported** [94]). The same MOIS-linked operator, tracked by ATT&CK as VOID MANTICORE, is tied by Check Point to the Handala persona that claimed the March 2026 Stryker attack (**Confirmed** [6] [76]).
- **Operational technology is a target in its own right (Inferred).** CyberAv3ngers, which Treasury and Rewards for Justice tie to the IRGC Cyber-Electronic Command, compromised at least 75 Unitronics PLC and HMI devices from November 2023 (**Confirmed** [23] [40] [51]); in 2026 an Iranian-affiliated actor manipulated project files and HMI displays on Rockwell, Schneider and Siemens PLCs and, in one case, disabled shutdown and alarm logic (**Confirmed** [29]).
- **Healthcare is hit mostly through shared infrastructure (Inferred).** The authors of AA21-321A assess the actors exploit known vulnerabilities rather than choosing sectors, and they reached the children's hospital's networks by exploiting a Fortigate appliance (**Confirmed** [17]). Research and pharmaceutical targeting is a vendor-reported theme: TA453 against medical researchers, APT42 against pharmaceutical firms in March 2020, Peach Sandstorm against pharmaceutical targets in 2023 (**Confirmed** [79] [64] [71]).
- **The library's data is a little behind ATT&CK in two places.** The live VOID MANTICORE page lists 75 technique rows against the library's 63, and the live CyberAv3ngers page 19 ICS rows and three software entries against the library's 17 and none (**Confirmed** for the counts [6] [12] [1]; the cause is **Inferred**).

Top recommendations

1. Patch every internet-facing VPN, firewall, mail server and remote-access gateway on the KEV list first, then hunt on the ones that were exposed. AA24-241A says patching alone may not be enough on a device that was vulnerable; AA22-320A says to treat VMware Horizon servers not patched promptly as compromised; AA20-259A says to rebuild compromised NetScaler devices (**Confirmed** [24] [21] [15]).
2. Treat identity noise as intrusion. Alert on many failed logins, one source IP across many accounts, impossible travel, MFA push floods and MFA device registrations from unexpected devices, and move administrators to phishing-resistant MFA (**Confirmed** guidance [25]). Pair this with the library's Multi-factor Authentication and User Geolocation Logon Pattern Analysis countermeasures.
3. Get controllers and building systems off the internet and off default passwords. Disconnect OT and ICS assets from the public internet, put any remote access behind a VPN or gateway with MFA, change default

ports and passwords, keep PLCs in run mode, and keep tested backups of controller logic (**Confirmed** guidance [23] [27]). In hospitals, start with HVAC, water, life-safety and building-automation systems, as the American Hospital Association advises (**Reported** [82]).

4. Watch for the access broker's toolkit. Unexpected AnyDesk, MeshCentral, Ligolo and ngrok, PowerShell Web Access enabled on servers, new local and domain accounts, new scheduled tasks and processes reading LSASS are the leads the advisories publish (**Confirmed** [24] [21]).
5. Protect device-management and identity consoles as tier-zero systems. The Stryker reporting points to a remote-wipe function in a management console, and the company's filing found no ransomware or malware (**Reported** [93]; **Confirmed** [59]; the link between them is **Inferred**). Limit who can issue fleet-wide wipe or reset actions, require approval for bulk actions, and log them to a place the console's own administrators cannot change (**Inferred** recommendation).
6. Plan for BitLocker abuse. Back up BitLocker recovery keys offline and alert on unexpected BitLocker activation (**Confirmed** guidance [19]).

Scope and method

This study covers activity that a government, a security vendor or MITRE ATT&CK attributes to the Iranian state, to one of its services, or to a contractor or front company working for them, from the 2011 to 2013 DDoS campaign against US banks through the reporting available on 3 November 2026. It draws on five kinds of source: US and allied government advisories, sanctions releases, court records and FBI wanted notices; security vendor reports; the MITRE ATT&CK pages for the groups and campaigns; the CISA Known Exploited Vulnerabilities (KEV) catalog; and the TeamStarWolf library's own data files at commit `e85b559` (ATT&CK v20.0). Every source in the Sources table was opened on 3 November 2026.

Each claim carries one of four grades.

Grade	Meaning
Confirmed	The primary source that published the fact (the agency, court, vendor or MITRE page) was opened on 3 November 2026 and says it. For an attribution, Confirmed means the source makes that attribution at the confidence it states; it does not upgrade that confidence
Reported	Only a secondary source (press, a trade association, or a scanned filing whose text could not be read) was available
Demonstrated	A public reproduction exists. No claim in this study rests on one
Inferred	This study's own reasoning from Confirmed facts, labelled where it appears

This study grades by source type (primary opened versus secondary only), the scheme the newer case studies use; under the case-studies README's original scheme a single vendor's report would read as Reported.

Four rules govern the names. The study says which name each source uses ("Microsoft's Peach Sandstorm", "Mandiant's APT42") instead of merging labels, and gives the ATT&CK object alongside where one exists. A source's own hedge travels with its claim ("assessed", "moderate confidence", "suspected"). Criminals who happen to be Iranian are kept apart from state actors: the SamSam ransomware operators were indicted as

criminals, and neither the FBI nor Treasury alleges a government tie (**Confirmed** [45] [35]). And tradecraft is described at the level the advisories use, by technique name and ATT&CK ID, with no commands, payload strings or exploit details.

Other studies in this series hold neighboring material. Iranian exploitation of Log4Shell appears here only as part of the AA22-257A and AA22-320A record; the vulnerability itself is the subject of the Log4Shell case study. The operation against Iran's enrichment program is the subject of the Stuxnet case study. The Change Healthcare case study covers the 2024 ALPHV/BlackCat attack on Change Healthcare; AA24-241A names ALPHV among the affiliates Pioneer Kitten worked with, but no source in this study links Pioneer Kitten to that attack. The ICS attacks and advisories case study is the home for the 2024 to 2026 operational technology incidents, including the IOCONTROL reporting and the AA26-097A PLC campaign; this study keeps the actor's dated record and summarizes those events. The 2021 Exchange exploitation that AA21-321A describes is set in its wider context in the Exchange Marauder case study, and the Iranian events of 2019 and 2020 sit beside the year's other intrusions in the 2019 and 2020 chronicles. POLONIUM, which the other state-linked threat actors study leaves to this record, is covered here; the China, Russia and United States studies hold the other national records in this series.

Some primaries could not be opened on 3 November 2026 and are not cited as opened: the DOJ press releases for the September 2024 campaign hack-and-leak charges, the March 2026 Handala domain seizures and the August 2026 expanded Mabna indictment; the Treasury release of 24 August 2026; the HHS HC3 brief "Iranian Threat Actors and Healthcare" (its hhs.gov PDF returned 403); the DNI's 2015 testimony text; and Israel's INCD notice on Ziv Hospital. The DOJ seizure warrant itself was opened, but it is a scanned PDF with no text layer, so its contents are graded from press reporting. Where this study relies on reporting about one of these, the claim says so.

Part 1: the actors and their sponsors

What ATT&CK records

In the library's ATT&CK v20.0 data, 21 groups carry an ATT&CK description that attributes them to Iran, the Iranian government, the IRGC or MOIS, or, for POLONIUM, to coordination with MOIS actors (**Confirmed** [1]). Nineteen are Enterprise groups, each with a generated library page; CyberAv3ngers (`G1027`) exists only in ATT&CK for ICS and UNC788 (`G1029`) only in ATT&CK for Mobile, and the library generates no page for either (**Confirmed** [1]). Inside the library, CyberAv3ngers can be reached through its tool IOControl (`S9046`) and the ICS ATT&CK Atlas.

Three keyword hits were read and left out: Strider (`G0041`), because Iran appears as a victim country; Ferocious Kitten (`G0137`), because it targets Persian-speaking people inside Iran and ATT&CK makes no state attribution; and HEXANE (`G1001`), because its own description makes no Iran attribution and it is named as Iran-nexus only in the narrative of the HomeLand Justice campaign (**Confirmed** [1] [13]).

ATT&CK group objects carry no first-seen or last-seen fields, so the "active since" column below comes from each description's own wording (**Confirmed** [1]). Technique and software counts are the library's direct group relationships; the live ATT&CK pages for some groups also roll up campaign relationships (see Part 9).

ATT&CK group	Other names in ATT&CK v20.0	Sponsor wording in ATT&CK	Active since (ATT&CK)	Techniques / software (library)
G0059 Magic Hound	TA453, COBALT ILLUSION, Charming Kitten, ITG18, Phosphorus, Newscaster, APT35, Mint Sandstorm	Iranian-sponsored, likely acting on behalf of the IRGC	2014	78 / 13
G0049 OilRig	COBALT GYPSY, IRN2, APT34, Helix Kitten, Evasive Serpens, Hazel Sandstorm, EUROPIUM, ITG13, Earth Simnavaz, Crambus, TA452	Suspected Iranian group working on behalf of the Iranian government; no service named	2014	76 / 30 (+6 ICS)
G0069 MuddyWater	Earth Vetala, MERCURY, Static Kitten, Seedworm, TEMP.Zagros, Mango Sandstorm, TA450, MuddyKrill	Subordinate element of MOIS	2017	68 / 21
G1059 NIMBUS MANTICORE	UNC1549, TA455, COBALT SMOKEY	State-sponsored espionage group associated with the IRGC	2021	65 / 13
G1055 VOID MANTICORE	COBALT MYSTIQUE, Handala Hack, Homeland Justice, Karma, Karmabelow80, BANISHED KITTEN, Red Sandstorm	Operates on behalf of MOIS	mid-2022	63 / 0
G0087 APT39	ITG07, Chafer, Remix Kitten	MOIS, through the front company Rana Intelligence Computing	2014	53 / 11
G0117 Fox Kitten	UNC757, Parisite, Pioneer Kitten, RUBIDIUM, Lemon Sandstorm	Suspected nexus to the Iranian government	2017	41 / 5
G1044 APT42	none listed	Iranian-sponsored; tracked separately from Magic Hound by the originating vendor	2015	33 / 2
G0064 APT33	HOLMIUM, Elfin, Peach Sandstorm	Suspected Iranian group; no service named	2013	31 / 16 (+3 ICS)
G1030 Agrius	Pink Sandstorm, AMERICIUM, Agonizing Serpens, BlackShadow	Public reporting links it to MOIS	2020	22 / 9
G1012 CURIUM	Crimson Sandstorm, TA456, Tortoise Shell, Yellow Liderc	Iranian; no service named	see page	19 / 1
G0077 Leafminer	Raspite	Iranian; no service named	see page	17 / 4
G0122 Silent Librarian	TA407, COBALT DICKENS	Affiliated with the Mabna Institute, acting for the IRGC	2013	13 / 0
G1009 Moses Staff	DEV-0500, Marigold Sandstorm	Suspected Iranian; no service named	see page	12 / 4
G0052 CopyKittens	none listed	Iranian; no service named	see page	8 / 4
G1005 POLONIUM	Plaid Rain	Lebanon-based, coordinating with MOIS actors	see page	7 / 2
G0130 Ajax Security Team	Operation Woolen-Goldfish, AjaxTM, Rocket Kitten, Flying Kitten, Operation Saffron Rose	Iranian; no service named	2010	6 / 2
G0003 Cleaver	Threat Group 2889, TG-2889	Iranian; no service named	see page	5 / 4

ATT&CK group	Other names in ATT&CK v20.0	Sponsor wording in ATT&CK	Active since (ATT&CK)	Techniques / software (library)
G0043 Group5	none listed	Iran nexus suspected, attribution "not definite"	see page	4 / 2
G1027 CyberAv3ngers	Soldiers of Soloman	Suspected IRGC-affiliated	2020	17 ICS / 0
G1029 UNC788	none listed	Iranian; no service named	see page	1 Mobile / 1

Sources: [1] for the table; the sponsor wording for Magic Hound, OilRig, MuddyWater, NIMBUS MANTICORE, VOID MANTICORE, APT39, Fox Kitten, APT42, APT33, Agrius and CyberAv3ngers was also read on the live pages [2] to [12]; the alias lists are from the library data [1] (**Confirmed**).

How the sponsors split

Read side by side, ATT&CK's sponsor wording sorts the set three ways (**Confirmed** [1]):

- **IRGC**: Magic Hound (likely), Silent Librarian (through the Mabna Institute), NIMBUS MANTICORE, CyberAv3ngers (suspected).
- **MOIS**: MuddyWater, APT39, VOID MANTICORE, Agrius (through public reporting), and POLONIUM as a Lebanon-based group that coordinates with MOIS actors.
- **Iran, with no service named**: OilRig, APT33, CopyKittens, Leafminer, Fox Kitten, Cleaver, Group5 (not definite), Ajax Security Team, Moses Staff (suspected), CURIUM, APT42 and UNC788.

Vendors go further than ATT&CK in places. Mandiant estimates with moderate confidence that APT42 operates for the IRGC Intelligence Organization, and Check Point says Educated Manticore (its name for the APT42 and Charming Kitten activity) is believed to work for the IRGC-IO (**Confirmed** [64] [77]). Microsoft assessed with moderate confidence that the initial-access actors behind the 2022 Albania attack are linked to EUROPIUM, which it says is publicly linked to MOIS; ATT&CK lists EUROPIUM as an OilRig alias but names no service for OilRig (**Confirmed** [60] [1] [3]). Google's Threat Analysis Group calls APT42 IRGC-associated (**Confirmed** [74]).

The contractors and fronts on the government record

The US government record puts names to the organizations that did the work. None of these are ATT&CK objects; the ATT&CK group each is closest to is given where a primary draws the link.

Organization	Service	What the record says	Closest ATT&CK object	Sources
ITSecTeam and Mersad	IRGC and the Iranian Government	Seven managers or employees indicted for DDoS attacks on the US financial sector, 2011 to 2013; ATT&CK says Havij is distributed by an Iranian security company named ITSecTeam (that it is the same company is Inferred)	Ajax Security Team uses Havij (S0224) (Inferred link)	[43] [1]

Organization	Service	What the record says	Closest ATT&CK object	Sources
Mabna Institute	IRGC	Contractor working at the behest of the IRGC; data theft from about 144 US universities, 176 universities in 21 other countries, five US government agencies and 36 US companies (2018 figures)	Silent Librarian G0122	[44] [34] [1]
Rana Intelligence Computing Company	MOIS	Front company for APT39; designated with APT39 and 45 individuals in 2020	APT39 G0087	[36] [7]
Net Peygard Samavat, renamed Emennet Pasargad, operating in 2024 as Aria Sepehr Ayandehsazan	IRGC (Net Peygard designated in 2019 for supporting the IRGC-EWCD)	2020 voter-intimidation operation; 2023 Charlie Hebdo leak (Microsoft: NEPTUNIUM); 2024 Olympics display compromise (FBI: Cotton Sandstorm)	none	[37] [70] [30]
Najee Technology Hooshmand Fater LLC and Afkar System Yazd Company	IRGC	Ransomware and BitLocker extortion; Treasury says the activity is partly attributable to intrusion sets private firms call APT35 and Charming Kitten	Magic Hound G0059 (through the APT35 alias)	[19] [39]
Danesh Novin Sahand	Government of Iran (FBI assessment)	Cover company used by Pioneer Kitten	Fox Kitten G0117	[24]
IRGC Cyber-Electronic Command, including its head Hamid Reza Lashgarian	IRGC	Six officials designated over the Unitronics PLC defacements	CyberAv3ngers G1027	[40] [51]
Mehrsam Andisheh Saz Nik and Dadeh Afzar Arman	IRGC Cyber-Electronic Command	Multi-year spear-phishing and malware campaign against more than a dozen US companies and government entities	none named	[41]

The link from Najee and Afkar to Magic Hound rests on Treasury's statement about APT35 and Charming Kitten and on ATT&CK listing both names as Magic Hound aliases; Treasury does not name the ATT&CK object (**Inferred** from [39] [1]).

Part 2: the operational record

One row per dated event, report or government action, oldest first. "Actor" is the name the source uses; the ATT&CK object follows where one exists. A report's publication date is given where the activity itself is not dated more precisely.

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2010	Earliest activity in the ATT&CK set	Ajax Security Team, G0130	Confirmed	[1]
2011 to 2013	Coordinated DDoS attacks on the US financial sector	ITSecTeam and Mersad staff, working for the Iranian Government including the IRGC	Confirmed	[43]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2012	Shamoon wiper first used, by a group calling itself Cutting Sword of Justice; Unit 42 later describes the 2012 target as a Saudi energy company	Shamoon S0140	Confirmed	[1] [67]
2013	Earliest activity for APT33 and Silent Librarian; the 2026 superseding Mabna indictment covers intrusions from 2013	APT33 G0064; Silent Librarian G0122	Confirmed (ATT&CK); Reported (2026 indictment)	[1] [88]
2014-02	Attack on Las Vegas Sands; DNI Clapper told the Senate Armed Services Committee on 2015-02-26 that Iran was behind it	not named	Reported	[96]
2015 (fall) to 2016-05	Helminth backdoor delivered by Excel lures, 4 to 12 May 2016, against Saudi financial and technology organizations, linked to a fall 2015 wave; Unit 42 calls the Iranian-operator indicators compelling but easily falsified	Unit 42's OilRig, G0049; Helminth S0170	Confirmed	[66]
2015-12 to 2018	SamSam ransomware: more than 6 million dollars from victims in critical infrastructure, healthcare, transportation and government; JBoss exploitation (early 2016), then RDP brute force and stolen credentials	Iran-based criminals; no government tie alleged	Confirmed	[45] [32] [35]
2016-01-21	SDNY indictment of seven ITSecTeam and Mersad employees (date as the FBI record gives it)	as above	Confirmed	[43]
mid-2016 to early 2017	Compromise of a US aerospace organization; targeting of a Saudi aviation conglomerate and a South Korean petrochemical firm; Mandiant had not seen APT33 run destructive operations (report 2017-09-20)	Mandiant's APT33, G0064	Confirmed	[61]
2016-11-17	Shamoon 2 Disttrack samples set to wipe on this date against at least one Saudi organization; Unit 42 does not attribute it to Iran	Shamoon S0140	Confirmed	[67]
2017-11	CVE-2017-11882 used against a Middle East government organization less than a week after the 14 November 2017 patch (report 2017-12-07)	Mandiant's APT34, loosely aligned with OilRig, G0049	Confirmed	[62]
2018-03-23	Nine Mabna Institute members announced as indicted (SDNY, 2018-02-07); Treasury sanctions the Institute the same day	Mabna Institute (IRGC); Silent Librarian G0122	Confirmed	[44] [34]
2018-11-26 to 2018-12-03	SamSam indictment in D.N.J. (2018-11-26), Treasury designation of two men who converted ransom payments (2018-11-28), CISA alert AA18-337A (2018-12-03)	Iran-based criminals	Confirmed	[45] [35] [32]
2019-01-29	Mandiant names APT39, tracked since November 2014, and assesses with moderate confidence that it supports Iranian national interests, prioritizing telecom and travel	Mandiant's APT39, G0087	Confirmed	[63]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2019-02-13	Former US Air Force intelligence specialist Monica Witt charged with espionage for Iran; four Iranians charged with a cyber campaign against US counterintelligence personnel	not named	Confirmed	[46]
2019-08 to 2019-09	More than 2,700 attempts to identify accounts and attacks on 241, including accounts tied to a US presidential campaign; four unrelated accounts compromised (report 2019-10-04)	Microsoft's Phosphorus, G0059	Confirmed	[69]
2020-03	Targeting of the pharmaceutical sector at the onset of COVID-19	Mandiant's APT42, G1044	Confirmed	[64]
2020-09-15	AA20-259A: an Iran-based actor exploits Pulse Secure, Citrix and F5 flaws against US IT, government, healthcare, financial, insurance and media organizations	Pioneer Kitten / UNC757, G0117	Confirmed	[15]
2020-09-17	Treasury designates APT39, Rana Intelligence Computing and 45 individuals for a campaign against dissidents, journalists and about 15 US companies, mainly travel, across more than 30 countries	APT39, G0087	Confirmed	[36]
2020-09-20 to 2020-10-20	US state election websites scanned and exploited (20 September to 17 October); voter registration data obtained in at least one state; voter-intimidation emails sent on 20 October (AA20-304A, 2020-10-30)	an Iranian APT (AA20-304A); Treasury's 2021 release describes matching activity by Emennet Pasargad (the match is Inferred)	Confirmed	[16] [37]
late 2020	Pay2Key hack-and-leak, which the FBI assesses was an information operation against Israeli cyber infrastructure, not a ransom operation	Pioneer Kitten; Pay2Key S0556	Confirmed	[24]
late 2020 to mid-2022	Watering hole and fake login pages, one spoofing Pfizer, against Israeli government, shipping, energy, aviation and healthcare organizations (report 2022-08-17)	Mandiant's UNC3890 (Iran-linked, moderate confidence); campaign C0010	Confirmed	[65] [1]
2020-12	Credential phishing of fewer than 25 senior staff at US and Israeli medical research organizations specializing in genetics, oncology and neurology (report 2021-03-30)	Proofpoint's TA453, G0059	Confirmed	[79]
2021-02 to 2021-10	A New Jersey municipality (February), small businesses hit with BitLocker ransom (March and April), transportation, emergency services, education and healthcare practices (June to August), a rural electric utility (October)	IRGC-affiliated group later designated as Najee Technology and Afkar System Yazd	Confirmed	[39]
2021-03	Fortinet exploitation begins	AA21-321A actors	Confirmed	[17]
2021-05	HomeLand Justice campaign first seen; initial access to Albanian government networks about 14 months before July 2022, through SharePoint CVE-2019-0604	VOID MANTICORE, C0038	Confirmed	[13] [20]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2021-06	A Fortigate appliance exploited to reach environmental control networks of a US hospital specializing in healthcare for children; Treasury says the hospital was notified before patient care was affected	AA21-321A actors; IRGC-affiliated per Treasury	Confirmed	[17] [39]
summer 2021	Hackers sponsored by the Iranian government went after Boston Children's Hospital; an intelligence partner tipped the FBI, which notified the hospital, and the threat was mitigated. Press reports the FBI contacted the hospital on 2021-08-03	not named	Confirmed (FBI); Reported (date)	[50] [95]
2021	Outer Space campaign	OilRig, C0042	Confirmed	[1]
2021-10	Exchange ProxyShell exploitation begins	AA21-321A actors	Confirmed	[17]
2021-11-17	AA21-321A (FBI, CISA, ACSC, NCSC): an APT assessed as associated with the government of Iran targets the Transportation and Healthcare and Public Health sectors	AA21-321A actors	Confirmed	[17]
2021-11-18	Treasury designates Emennet Pasargad and six associates for the 2020 election operation	Emennet Pasargad	Confirmed	[37]
2021-12 and 2022-02	Log4j exploitation; BitLocker ransom against a US police department and a regional transport company in December 2021	IRGC-affiliated actors (AA22-257A)	Confirmed	[19]
2022-02	Log4Shell exploited on an unpatched VMware Horizon server at a Federal Civilian Executive Branch agency, followed by XMRig, Mimikatz, PsExec and ngrok (AA22-320A, 2022-11-16)	likely Iranian government-sponsored actors	Confirmed	[21]
2022	Juicy Mix campaign	OilRig, C0044	Confirmed	[1]
2022-02-24	AA22-055A (FBI, CISA, CNMF, NCSC): MuddyWater is a subordinate element of MOIS, running campaigns for MOIS since about 2018 against telecom, defense, local government and oil and gas	MuddyWater, G0069	Confirmed	[18]
2022-06-01	FBI Director Wray describes the thwarted Boston Children's Hospital attack	as above	Confirmed	[50]
2022-07-15 to 2022-07-18	Destructive attack on the Government of Albania begins on 15 July: a ransomware-style encryptor then the ZeroCleare wiper; HomeLand Justice claims it on 18 July; medical appointment booking among the disrupted services	Iranian state actors (US, UK, Microsoft); VOID MANTICORE G1055; ZeroCleare S1151	Confirmed	[20] [52] [60]
2022-08-10	D.N.J. indictment of Mansour Ahmadi, Ahmad Khatibi Aghda and Amir Hossein Nickaein Ravari for intrusion, encryption and extortion from October 2020 to August 2022; Rewards for Justice offers up to 10 million dollars	the IRGC-affiliated Najee Technology and Afkar System Yazd group, per Treasury	Confirmed	[47] [39]
2022-08-17	Mandiant publishes UNC3890	UNC3890; C0010	Confirmed	[65]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2022-09-06	Mandiant estimates with moderate confidence that APT42 operates for the IRGC Intelligence Organization, with more than 30 confirmed operations since early 2015	APT42, G1044	Confirmed	[64]
2022-09-07	UK condemns the Iranian state for the Albania attack; NCSC assesses Iranian state-linked actors almost certainly responsible. Albania severs diplomatic relations and gives Iranian diplomats 24 hours to leave	as above	Confirmed (UK); Reported (Albania)	[52] [53] [94]
2022-09-08	EU High Representative expresses solidarity with Albania, together with NATO, and condemns the attack without naming Iran; Microsoft attributes it with high confidence to Iranian government-sponsored actors	as above	Confirmed	[54] [60]
2022-09-09	Treasury designates MOIS and Minister Esmail Khatib over the Albania attack and leaks, and names MuddyWater as a subordinate element of MOIS	MOIS	Confirmed	[38]
2022-09	Second wave of attacks on Albanian government networks	HomeLand Justice actors	Confirmed	[20] [13]
2022-09-14	AA22-257A re-assesses the AA21-321A actors as IRGC-affiliated, operating under Najee Technology and Afkar System Yazd; Treasury sanctions ten individuals and the two companies in a joint action with DOJ and State	IRGC-affiliated actors	Confirmed	[19] [39]
2022-09-20	NCSC, with US, Australian and Canadian agencies, publicly exposes IRGC-affiliated actors exploiting Fortinet, Exchange and Log4j for ransomware	as above	Confirmed	[55]
2022-09-21	AA22-264A attributes the Albania attack to Iranian state cyber actors using the HomeLand Justice persona	as above	Confirmed	[20]
2022-11-03	HHS HC3 publishes the TLP:CLEAR brief "Iranian Threat Actors and Healthcare"; trade coverage says it names Pioneer Kitten, Magic Kitten, Infy and UNC3890	several	Reported	[81]
2022-11-16	AA22-320A on the FCEB Log4Shell intrusion	as above	Confirmed	[21]
2023-01-04	"Holy Souls" leak of Charlie Hebdo customer data (200,000 records claimed); Microsoft believes it was an Iranian government response (report 2023-02-02)	Microsoft's NEPTUNIUM, Emennet Pasargad	Confirmed	[70]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2023-02 to 2023-07	Password spraying against thousands of organizations, focused on satellite, defense and, to a lesser extent, pharmaceutical targets; exploitation of CVE-2022-47966 and CVE-2022-26134 in a subset of intrusions (report 2023-09-14)	Microsoft's Peach Sandstorm, G0064	Confirmed	[71]
2023-10 onward	Brute force, password spraying and MFA push bombing against healthcare and public health, government, IT, engineering and energy organizations; access likely sold (AA24-290A)	Iranian actors (not named)	Confirmed	[25]
2023-10 to 2023-12	After 7 October, 43 percent of the Iranian nation-state activity Microsoft tracked targeted Israel; Israel's INCD records hospital targeting as a main trend of the war's cyber dimension	various	Confirmed	[72] [58]
2023-11	Tailored social engineering against Middle East affairs researchers in Belgium, France, Gaza, Israel, the UK and the US, with the MediaPl and MischiefTut backdoors (report 2024-01-17)	Microsoft's Mint Sandstorm, G0059	Confirmed	[73]
2023-11-22 to 2024-01	At least 75 Unitronics PLC and HMI devices compromised, 34 of them in US water and wastewater; Microsoft says a PLC at a Pennsylvania water authority was taken offline on 25 November	CyberAv3ngers, G1027	Confirmed	[23] [72]
2023-11-28	CISA alert on exploitation of Unitronics PLCs in water and wastewater, through weak passwords and internet exposure	as above	Confirmed	[22]
2023-12-01	AA23-335A attributes the CyberAv3ngers persona to IRGC-affiliated actors; healthcare listed among the sectors that use these PLCs	as above	Confirmed	[23]
2023-12-11	CISA adds Unitronics CVE-2023-6448 to KEV	as above	Confirmed	[33]
2023-12-18	INCD attributes an attempted attack on Ziv Hospital to Iran and Hezbollah, as footnoted by Israel's Foreign Ministry (the INCD notice itself not opened; Israel is a party to the conflict)	not named	Confirmed (that the Ministry says so)	[57]
2024-01-12	INCD Alert 1676: groups working with Iranian intelligence attempted operations against Israeli water, energy and IT infrastructure (as cited by Israel's Foreign Ministry)	not named	Confirmed (that the Ministry says so)	[57]
2024-02-02	Treasury designates six IRGC Cyber-Electronic Command officials, including its head, over the PLC defacements	IRGC-CEC	Confirmed	[40]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2024-04-23	Treasury designates IRGC-CEC fronts Mehrsam Andisheh Saz Nik and Dadeh Afzar Arman and four individuals; DOJ unseals an indictment of four people the same day, including Alireza Shafie Nasab (as Treasury states)	IRGC-CEC	Confirmed	[41]
2024-05 to 2024-06	Personal email of roughly a dozen people tied to the Biden and Trump campaigns targeted; the US and Israel about 60 percent of known targeting over six months (report 2024-08-14)	Google TAG's APT42, G1044	Confirmed	[74]
2024-05 onward	Accounts of officials and advisors to a 2024 presidential campaign compromised and the data leaked to media	IRGC members including Masoud Jalili	Confirmed	[42]
2024-07	Compromise of a French dynamic display provider during the Summer Olympics, and harvesting of IP camera content (advisory 2024-10-30)	Emennet Pasargad as Aria Sepehr Ayandehsazan (Cotton Sandstorm)	Confirmed	[30]
2024-08-19	ODNI, FBI and CISA attribute the reported compromise of the Trump campaign to Iran	Iran	Confirmed	[31]
2024-08-28	AA24-241A on Pioneer Kitten's access brokering for ransomware affiliates, 2017 through August 2024	Pioneer Kitten, G0117	Confirmed	[24]
2024-09-27	Masoud Jalili, Yasar Balaghi and Seyyed Ali Aghamiri charged in D.D.C.; Treasury designates seven Iranians including Jalili	IRGC-affiliated	Confirmed	[48] [42] [89]
2024-10-16	AA24-290A on brute force and MFA push bombing (FBI, CISA, NSA, CSE, AFP, ASD's ACSC)	Iranian actors	Confirmed	[25]
2024-12-18	AA23-335A update: custom ladder logic, renamed devices, disabled upload and download functions, changed default ports, NCSC-observed PLC targeting including in the UK	CyberAv3ngers	Confirmed	[23]
2025-06 (mid)	Spear-phishing wave against Israeli journalists, cyber experts and computer science professors after the Iran-Israel conflict (report 2025-06-25)	Check Point's Educated Manticore, G1044 and G0059	Confirmed	[77]
2025-06-27	Health-ISAC warns of rising Iranian threats to healthcare and of state actors posing as hacktivists	various	Reported	[84]
2025-06-30	CISA, FBI, DC3 and NSA fact sheet: no sign of a coordinated campaign in the US yet, but heightened risk, especially for Defense Industrial Base firms with Israeli ties	various	Confirmed	[26] [27] [28]
2025-12-02	MuddyViper backdoor and a Snake-game-themed loader against Israeli critical infrastructure and one Egyptian target	ESET's MuddyWater, G0069; MuddyViper S9032	Confirmed	[78]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
late 2025 to early 2026	Starlink used for command and control	MuddyWater, G0069	Confirmed	[4]
2026-02 (late)	Pay2Key ransomware intrusion at a US healthcare organization; no evidence of exfiltration	Beazley Security's Pay2Key (Iranian government-linked)	Confirmed	[80]
2026-02-28	US-Israel strikes mark the start of an Iranian retaliatory campaign; a 47-day near-total Iranian internet outage follows; state actors moved to Starlink (high confidence)	Iranian state actors (Unit 42)	Confirmed	[68]
2026-03-02	NCSC: likely no significant change in the direct threat to the UK, but almost certainly a heightened indirect risk for organizations with a Middle East presence	Iran-linked actors (NCSC assessment)	Confirmed	[56]
2026-03-03	FBI Cyber Division reminds critical infrastructure to apply the June 2025 mitigations; AHA tells hospitals to inventory internet-facing OT	FBI, AHA	Reported	[82]
2026-03 onward	Tampering with internet-exposed Rockwell/Allen-Bradley, Schneider Electric and Siemens PLCs in government facilities, water and energy; project files and HMI displays manipulated; in one case shutdown and alarm logic disabled	an Iranian-affiliated APT (AA26-097A); Unit 42's CL-STA-1128 (Cyber Av3ngers) targeting Rockwell OT	Confirmed	[29] [68]
2026-03-11	Stryker identifies a cybersecurity incident causing a global disruption to its Microsoft environment, with no indication of ransomware or malware; Handala claims a wiper attack	Handala, which Check Point ties to Void Manticore, G1055	Confirmed (8-K, Check Point); Reported (claims)	[59] [76] [93]
2026-03-19	DOJ seizes four domains under a D. Md. warrant (case 1:26-mj-00683-CDA); reporting says DOJ attributes the Handala, Homeland Justice and Karma Below personas to MOIS and that Maryland hospitals and emergency services were affected	MOIS	Confirmed (filing exists); Reported (contents)	[49] [90] [91] [92]
2026-04-07	AA26-097A published (revised 2026-07-22)	as above	Confirmed	[29]
2026-04-20	ATT&CK creates VOID MANTICORE, whose description names the March 2026 Stryker attack under the Handala Hack persona	G1055	Confirmed	[6]
2026-08-18	Superseding indictment of 17 Mabna Institute members for intrusions for the IRGC since 2013: 144 US universities, 42 US companies, at least five agencies, more than 31 TB stolen; Behzad Mesri named in the HBO intrusion and extortion	Mabna Institute	Reported	[87] [88]
2026-08-24	Treasury sanctions five Iranian nationals over a hacking operation reported as directed by MOIS; four also charged in the expanded Mabna case	reported as MOIS	Reported	[86]

Date	Event	Actor (source's name; ATT&CK)	Grade	Sources
2026-10-01	ATT&CK creates NIMBUS MANTICORE, an IRGC-associated espionage group active since 2021 against aviation, aerospace, telecom, defense and IT organizations	G1059	Confirmed	[5]

Reading the record by era

Read together, the US record shows a shift in purpose over fifteen years (**Inferred** from the rows above; no single document states it). From 2011 to about 2020 the visible work is espionage against people the state watches, by MOIS (APT39's travel and telecom collection) and by the IRGC (Magic Hound, likely acting for the IRGC, against journalists and officials), plus IRGC contractor data theft (Mabna's universities), with the bank DDoS campaign as an early disruptive outlier. From about 2020 to 2024 the edge-device clusters turn to access brokering and ransomware, either alongside criminal affiliates (Pioneer Kitten) or with their own BitLocker extortion (Najee and Afkar). From late 2023 the IRGC Cyber-Electronic Command and its CyberAv3ngers persona move into operational technology, and MOIS-linked operators run destructive wiper and hack-and-leak operations: Agrius against Israel from 2020, the HomeLand Justice persona against Albania in 2022, and in 2026 the Handala persona against a US medical technology company.

Where the sources disagree

- **When MuddyWater started.** ATT&CK says since at least 2017; AA22-055A says it has run campaigns for MOIS since about 2018 (**Confirmed** [4] [18]).
- **Whether the children's hospital is Boston Children's.** AA21-321A and Treasury place the access in June 2021; Director Wray describes the Boston Children's attempt as summer 2021, and press puts the FBI's call to the hospital on 3 August 2021. Neither primary says the two are the same event (**Inferred** that they may be).
- **Whether the 2026 PLC actor is CyberAv3ngers.** AA26-097A calls the actor Iranian-affiliated and says the agencies previously reported similar PLC activity by CyberAv3ngers (aka Shahid Kaveh Group), which it ties to the IRGC Cyber-Electronic Command. It does not state that the 2026 actor is CyberAv3ngers (**Confirmed** [29]). Unit 42 does use the Cyber Av3ngers name for its 2026 Rockwell cluster (**Confirmed** [68]).
- **How many Stryker devices were wiped.** Handala claimed more than 200,000; TechCrunch reports tens of thousands. Both are claims or press figures, and Stryker's filing gives no count (**Reported** [93] [91]; **Confirmed** [59]).
- **Who directed the August 2026 sanctioned group.** Press reports Treasury as saying MOIS directed the operation, while the Mabna case it overlaps is an IRGC case. The Treasury release could not be opened, so this study does not resolve it (**Reported** [86]).
- **Shamoon's author.** ATT&CK records the 2012 Shamoon use by a group calling itself Cutting Sword of Justice, which it describes as Iranian; Unit 42's 2016 Shamoon 2 report does not attribute the attack to Iran (**Confirmed** [1] [67]).
- **Albania and the EU.** The US, UK, Microsoft and Treasury name Iran or MOIS; the EU's declaration condemns the attack but names no state (**Confirmed** [20] [52] [60] [38] [54]).

Part 3: tooling

ATT&CK software attributed to the Iranian groups

The 19 Enterprise groups use 111 distinct ATT&CK software entries, every one with a generated library page (**Confirmed** [1]). The shared tools are mostly public or dual-use: Mimikatz (S0002, 8 groups), PsExec (S0029, 7), LaZagne (S0349, 4), Empire (S0363, 3), Net (S0039, 3) and ftp (S0095, 3). Two groups each use ngrok (S0508), CrackMapExec (S0488), ASPXSpy (S0073), DEADWOOD (S1134, APT33 and Agrius), PowerSploit (S0194), Pupy (S0192) and NBTscan (S0590) (**Confirmed** [1]).

Group	Group-specific families in ATT&CK (selection)	Sources
OilRig G0049	BONDUPDATER S0360, Helminth S0170, POWRUNER S0184, SideTwist S0610, RDAT S0495, Solar S1166, Mango S1169, and the wiper ZeroCleare S1151	[1]
MuddyWater G0069	POWERSTATS S0223, PowGoop S1046, Small Sieve S1035, STARWHALE S1037, Mori S1047, MuddyViper S9032, RustyWater S9037; remote-management tools ConnectWise S0591 and RemoteUtilities S0592	[1] [18] [78]
Magic Hound G0059	CharmPower S0674, PowerLess S1012	[1]
APT33 G0064	StoneDrill S0380, POWERTON S0371, TURNEDUP S0199	[1]
Agrius G1030	Apostle S1133, MultiLayer Wiper S1135, Moneybird S1137	[1]
Fox Kitten G0117	Pay2Key S0556, SystemBC S9001	[1]
Moses Staff G1009	PyDCrypt S1032, DCSrv S1033, StrifeWater S1034	[1]
APT42 G1044	NICECURL S1192, TAMECAT S1193	[1] [9]
NIMBUS MANTICORE G1059	MiniBike S9059, MiniBus S9060, MiniFast S9052 and ten more	[1] [5]
Ajax Security Team G0130	Havij S0224, an ITSecTeam tool	[1]
CyberAv3ngers G1027	IOControl S9046; the live group page also names three PLC programming tools as software, which the library data does not yet hold (see Part 9)	[1] [12]

AA22-055A names MuddyWater's PowGoop, Small Sieve, Canopy (Starwhale), Mori and POWERSTATS; each is an ATT&CK software object under that name, Canopy as an alias of STARWHALE (**Confirmed** [18] [1]). Rewards for Justice says IOCONTROL was used against vendors including Unitronics (**Confirmed** [51]).

Wipers and the destructive line

The Iranian record carries a recognizable wiper lineage. Shamoon (S0140) was first used in 2012; ZeroCleare (S1151), linked to OilRig in ATT&CK and a RawDisk-based wiper since at least 2019, followed a ransomware-style encryptor in the July 2022 Albania attack (**Confirmed** [1] [20]). The HomeLand Justice campaign also used CHIMNEYSWEEP (S1149) and ROADSWEEP (S1150) (**Confirmed** [13]). Check Point says Void Manticore's CI Wiper was first used against Albania in July 2022, and that Scarred Manticore hands off victims to Void Manticore for the destructive phase (**Confirmed** [75]). Agrius carries its own destructive tools: MultiLayer Wiper, and Apostle, which ATT&CK describes as both a wiper and, in later versions, ransomware (**Confirmed** [1]). Stryker may mark a change of method: a source told KrebsOnSecurity the attackers appear to have used Microsoft Intune's remote wipe, and the company's filing found no ransomware or malware (**Reported** [93]; **Confirmed** [59]; the fit between them is **Inferred**).

Remote access and tunneling

Remote-management and tunneling tools recur across the record. AA24-241A names AnyDesk, MeshCentral, Ligolo and ngrok; AA22-320A records ngrok after the Horizon compromise; MuddyWater's ATT&CK record includes ConnectWise and RemoteUtilities (**Confirmed** [24] [21] [1]). In 2026 Unit 42 assesses with high confidence that state actors moved command and control to Starlink during the Iranian internet outage, and ATT&CK records MuddyWater using Starlink for C2 (**Confirmed** [68] [4]).

Part 4: exploited vulnerabilities

The joint advisories name 20 distinct IT CVEs from 2020 to 2024. Nineteen are in the KEV catalog (version 2026.11.03), each marked as having known ransomware campaign use; CVE-2021-45105 is not in KEV. The Unitronics flaw CVE-2023-6448 was added on 11 December 2023 with ransomware use unknown (**Confirmed** [33], checked against the KEV JSON). CWE IDs below are the KEV catalog's own `cwes` field. Four more CVEs follow the joint-advisory rows: one from AA22-264A (the Albania attack) and three from vendor reports.

CVE	Product	Named in	KEV date added	CWE (KEV)
CVE-2019-11510	Pulse Connect Secure	AA20-259A [15]	2021-11-03	CWE-22
CVE-2019-11539	Pulse Connect Secure and Policy Secure	AA20-259A [15]	2021-11-03	CWE-78
CVE-2019-19781	Citrix ADC, Gateway and SD-WAN WANOP	AA20-259A [15], AA24-241A [24]	2021-11-03	CWE-22
CVE-2020-5902	F5 BIG-IP	AA20-259A [15]	2021-11-03	CWE-22
CVE-2018-13379	Fortinet FortiOS	AA21-321A [17], AA22-257A [19]	2021-11-03	CWE-22
CVE-2020-12812	Fortinet FortiOS	AA21-321A [17]	2021-11-03	CWE-178, CWE-287
CVE-2019-5591	Fortinet FortiOS	AA21-321A [17]	2021-11-03	CWE-306
CVE-2021-34473	Microsoft Exchange Server (ProxyShell)	AA21-321A [17], AA22-257A [19]	2021-11-03	CWE-918
CVE-2021-34523	Microsoft Exchange Server (ProxyShell)	AA22-257A [19]	2021-11-03	CWE-287

CVE	Product	Named in	KEV date added	CWE (KEV)
CVE-2021-31207	Microsoft Exchange Server (ProxyShell)	AA22-257A [19]	2021-11-03	CWE-20, CWE-434
CVE-2020-1472	Microsoft Netlogon	AA22-055A [18]	2021-11-03	CWE-330
CVE-2020-0688	Microsoft Exchange Server	AA22-055A [18]	2021-11-03	CWE-287
CVE-2021-44228	Apache Log4j2 (Log4Shell)	AA22-257A [19], AA22-320A [21]	2021-12-10	CWE-20, CWE-400, CWE-502
CVE-2021-45046	Apache Log4j2	AA22-257A [19]	2023-05-01	CWE-917
CVE-2021-45105	Apache Log4j2	AA22-257A [19]	not in KEV	not given
CVE-2023-3519	Citrix NetScaler ADC and Gateway	AA24-241A [24]	2023-07-19	CWE-94
CVE-2022-1388	F5 BIG-IP	AA24-241A [24]	2022-05-10	CWE-306
CVE-2024-21887	Ivanti Connect Secure and Policy Secure	AA24-241A [24]	2024-01-10	CWE-77
CVE-2024-3400	Palo Alto Networks PAN-OS	AA24-241A [24]	2024-04-12	CWE-20, CWE-77
CVE-2024-24919	Check Point Quantum Security Gateways	AA24-241A [24]	2024-05-30	CWE-200
CVE-2023-6448	Unitronics Vision PLC and HMI	KEV only (added 2023-12-11, after the CyberAv3ngers activity; no advisory here names the CVE, so the link to CyberAv3ngers is **Inferred**) [33]	2023-12-11	CWE-1188
CVE-2019-0604	Microsoft SharePoint	AA22-264A [20] (Albania initial access)	2021-11-03	CWE-20
CVE-2017-11882	Microsoft Office	Mandiant APT34 report [62]	2021-11-03	CWE-119
CVE-2022-47966	Zoho ManageEngine	Microsoft Peach Sandstorm report [71]	2023-01-23	not given
CVE-2022-26134	Atlassian Confluence	Microsoft Peach Sandstorm report [71]	2022-06-02	CWE-917

The pattern is old flaws in edge devices. Most of the CVEs were disclosed months or years before the advisory that names them, and the gap between disclosure and exploitation is not the point: the actors scan for appliances that were never patched, and AA21-321A's authors assess they exploit known vulnerabilities rather than choose sectors (**Confirmed** [17]; the reading is **Inferred**). The one fast case in the record is Mandiant's APT34 using CVE-2017-11882 less than a week after its patch (**Confirmed** [62]).

Part 5: targets by sector and geography

Geography. The US, Israel and the Gulf states dominate the record, with Albania as the clearest European case (**Inferred** from the rows below). ATT&CK places APT33 against the US, Saudi Arabia and South Korea, NIMBUS MANTICORE against the US, Australia, the Middle East and Europe, and VOID MANTICORE against Albania, Israel and the US (**Confirmed** [10] [5] [6]). MuddyWater's MOIS campaigns span Asia, Africa, Europe and North America (**Confirmed** [18]). Pioneer Kitten's victims abroad include Israel, Azerbaijan and the UAE (**Confirmed** [24]). AA22-257A lists targets in the US, Australia, Canada and the UK, and the NCSC judges Afkar System Yazd is targeting UK organizations (**Confirmed** [19]). Google's TAG found the US and Israel made up about 60 percent of APT42's known targeting over six months in 2024 (**Confirmed** [74]). After 7 October 2023, 43 percent of the Iranian nation-state activity Microsoft tracked targeted Israel (**Confirmed** [72]).

Sectors. The sector lists in the record fall into four groups (**Confirmed** for each listing; the grouping is **Inferred**):

Theme	Sectors	Example sources
People the state watches	dissidents, journalists, think tanks, academics, diaspora, officials, election campaigns, the travel and telecom data that track them	[2] [7] [36] [64] [69] [74] [42]
Strategic industry and research	aviation, aerospace, defense, satellite, energy, petrochemical, pharmaceutical and medical research, universities	[10] [61] [5] [71] [79] [44]
Whatever is exposed	IT, government, municipalities, transportation, police, education, finance, insurance, media, healthcare	[15] [17] [19] [24] [25]
Operational technology	water and wastewater, energy, government facilities, food and beverage, healthcare (as a user of the same PLCs)	[23] [27] [29] [51]

Part 6: the healthcare record

Someone running a hospital security program asks a narrower question than a threat-intelligence team does: which of these actors has touched healthcare, how, and on whose word. The table answers it. The US government record names the Healthcare and Public Health sector for four Iran-attributed clusters; research and pharmaceutical targeting rests on vendor reporting (**Inferred** summary of the rows).

Date	Healthcare target	Actor and attribution	Grade	Sources
2015-12 to 2018	Hospitals and medical centers among SamSam victims (press names LabCorp)	Iran-based criminals; no state tie alleged	Confirmed (FBI, Treasury); Reported (victim names)	[45] [35] [97]
2020-03	Pharmaceutical sector at the onset of COVID-19	Mandiant's APT42 (IRGC-IO, moderate confidence)	Confirmed	[64]
2020-09-15	Healthcare among the US sectors hit through Pulse Secure, Citrix and F5	Pioneer Kitten / UNC757	Confirmed	[15]
2017 to 2024-08	US healthcare facilities among the victims of the access broker	Pioneer Kitten (Government of Iran, FBI)	Confirmed	[24]

Date	Healthcare target	Actor and attribution	Grade	Sources
late 2020 to mid-2022	Israeli healthcare organizations; a fake login page spoofing Pfizer	Mandiant's UNC3890 (Iran-linked, moderate confidence; not tied to the IRGC)	Confirmed	[65] [1]
2020-12	Senior staff at US and Israeli medical research organizations in genetics, oncology and neurology	Proofpoint's TA453; Proofpoint could not tie it independently to the IRGC but says the targeting appears to support IRGC collection priorities	Confirmed	[79]
2021-06	Environmental control networks of a US children's hospital, reached through a Fortigate appliance	AA21-321A actors; IRGC-affiliated per Treasury	Confirmed	[17] [39]
2021-06 to 2021-08	Healthcare practices	the same IRGC-affiliated group	Confirmed	[39]
summer 2021	Boston Children's Hospital (attempt thwarted after an FBI warning)	hackers sponsored by the Iranian government (FBI)	Confirmed	[50]
2022-07	Albanian government services, including medical appointment booking	Iranian state (UK, US)	Confirmed	[52] [20]
2023-02 to 2023-07	Pharmaceutical targets, to a lesser extent than satellite and defense; healthcare among past Peach Sandstorm targets	Microsoft's Peach Sandstorm (APT33)	Confirmed	[71]
2023-10 to 2023-12	Israeli hospitals, which INCD says state attack groups treat as a legitimate target	various	Confirmed	[58]
2023-10 onward	Healthcare and public health among the brute-force and MFA push bombing victims	Iranian actors (AA24-290A)	Confirmed	[25]
2023-11 to 2024-01	Healthcare among sectors that use the targeted Unitronics PLCs; the June 2025 fact sheet says US victims included the HPH sector	CyberAv3ngers (IRGC-affiliated)	Confirmed	[23] [27]
2023-12-18	Attempted attack on Ziv Hospital, attributed by INCD to Iran and Hezbollah	as cited by Israel's Foreign Ministry	Confirmed (that the Ministry says so)	[57]
2026-02 (late)	Pay2Key ransomware at a US healthcare organization	Pay2Key (Iranian government-linked, Beazley Security)	Confirmed	[80]
2026-03	Handala's claimed targeting of Israeli healthcare	Unit 42	Confirmed (that Unit 42 records the claim)	[68]
2026-03-11	Stryker, a medical technology company; some hospitals suspended connections to Stryker systems; prosecutors say emergency medical services and hospitals in Maryland were affected	Handala (MOIS, per reporting on DOJ filings; Void Manticore, per Check Point)	Confirmed (8-K, Check Point); Reported (impact, DOJ attribution)	[59] [76] [90] [93]

Three readings follow. First, almost every confirmed US hospital case comes through an exposed edge device, a weak password or an internet-facing controller, the same front doors the actors use everywhere; the

record shows opportunism, not a hospital-specific campaign (**Inferred** from [17] [24] [25] [23]). Second, the research and pharmaceutical cases are espionage against people, through credential phishing and social engineering. Proofpoint says the medical-research targeting appears to support IRGC collection priorities, and Mandiant attributes the APT42 pharmaceutical targeting to an actor it assesses works for the IRGC-IO (**Confirmed** [79] [64]). Third, the supply chain is an exposure point (**Inferred**): in 2026 a medical technology vendor's outage reached hospitals, and in June 2025 Health-ISAC warned that healthcare supply chains and Israeli-based software vendors are exposure points (**Confirmed** [59]; **Reported** [84] [90]).

The HHS HC3 brief "Iranian Threat Actors and Healthcare" of 3 November 2022 is the sector's own summary, but its primary copy could not be opened; trade coverage says it names Pioneer Kitten, Magic Kitten, Infy and UNC3890 and recommends phishing training and network segmentation (**Reported** [81]). AHA's guidance in July 2025 and March 2026 tells hospitals to stay alert and to inventory internet-facing OT and ICS such as HVAC, water, life-safety and building automation (**Reported** [83] [82]).

Part 7: the government record

United States: indictments, seizures and rewards

Date	Court or agency	Defendants or target	Allegation	Sources
2016-01-21	SDNY	Seven ITSecTeam and Mersad managers or employees	DDoS on the US financial sector, 2011 to 2013, for the Iranian Government including the IRGC	[43]
2018-02-07 (announced 2018-03-23)	SDNY	Nine Mabna Institute members	Theft from universities, agencies and companies at the behest of the IRGC	[44]
2018-11-26	D.N.J.	Shah Mansouri and Shahi Savandi	SamSam ransomware; no government tie stated	[45]
2019-02-13	FBI record (court not given here)	Monica Witt and four Iranians	Espionage for Iran; cyber campaign against US counterintelligence personnel	[46]
2022-08-10	D.N.J.	Mansour Ahmadi, Ahmad Khatibi Aghda, Amir Hossein Nickaein Ravari	Intrusion, exfiltration, encryption and extortion against critical infrastructure, agencies and non-profits, October 2020 to August 2022; unsealed as part of the action against the IRGC-affiliated Najee Technology and Afkar System Yazd group, per Treasury	[47] [39]
2024-04-23	DOJ and FBI (as stated by Treasury)	Four individuals tied to IRGC-CEC fronts, including Alireza Shafie Nasab	Spear-phishing and malware against US companies and agencies	[41]
2024-09-27	D.D.C.	Masoud Jalili, Yasar Balaghi, Seyyed Ali Aghamiri	Targeting people tied to the 2024 presidential campaigns, officials, think tank staff, journalists and activists since 2019; hack-and-leak to media	[48] [89]
2026-03-19	D. Md., case 1:26-mj-00683-CDA	Four domains	Seizure warrant; reporting says the affidavit ties Handala, Homeland Justice and Karma Below to MOIS and matches the Stryker incident	[49] [90] [91] [92]
2026-08-18	DOJ (superseding indictment)	17 Mabna Institute members, including Behzad Mesri	Intrusions for the IRGC since 2013; HBO intrusion and extortion	[87] [88]

The FBI page for Ahmadi and his co-defendants and the Rewards for Justice page for CyberAv3ngers each advertise rewards of up to 10 million dollars (**Confirmed** [47] [51]).

United States: sanctions

Date	Release	Designated	Basis	Sources
2018-03-23	sm0332	Mabna Institute and individuals	IRGC outsourced intrusions to Mabna	[34]
2018-11-28	sm556	Two Iran-based men	Converting SamSam ransom payments; no claim the government directed SamSam	[35]
2020-09-17	sm1127	APT39, Rana Intelligence Computing, 45 individuals	Owned or controlled by MOIS	[36]
2021-11-18	jy0494	Emennet Pasargad and six individuals (E.O. 13848)	2020 election influence and intimidation	[37]
2022-09-09	jy0941	MOIS and Minister Esmail Khatib	Albania attack and leaks; MuddyWater named as a MOIS element	[38]
2022-09-14	jy0948	Ten individuals, Najee Technology, Afkar System Yazd	IRGC-affiliated ransomware; a children's hospital's SCADA access in June 2021	[39]
2024-02-02	jy2072	Six IRGC-CEC officials	Unitronics PLC defacements	[40]
2024-04-23	jy2292	Mehrsam Andisheh Saz Nik, Dadeh Afzar Arman, four individuals	Spear-phishing and malware, including malware testing aimed at job seekers with a focus on military veterans	[41]
2024-09-27	jy2621	Seven Iranians including IRGC member Masoud Jalili (E.O. 13848)	2024 campaign hack-and-leak	[42]
2026-08-24	not opened	Five Iranian nationals	A hacking operation reported as directed by MOIS	[86] (Reported)

United States: advisories and joint statements

Every advisory in the record, with its co-authors, is in the timeline in Part 2. The allied co-authorship is itself part of the record: Australia's ACSC and the UK NCSC co-signed AA21-321A; the NCSC, ACSC and Canada's CCCS co-signed AA22-257A; Israel's INCD, CCCS and the NCSC co-signed AA23-335A; Canada's CSE, the Australian Federal Police and ASD's ACSC co-signed AA24-290A (**Confirmed** [17] [19] [23] [25]). The FBI, Treasury and the INCD issued the October 2024 Emennet Pasargad advisory together (**Confirmed** [30]). On 19 August 2024 ODNI, FBI and CISA attributed the reported Trump campaign compromise to Iran (**Confirmed** [31]).

The United Kingdom, the EU, Albania and Israel

The UK government condemned Iran for the Albania attack on 7 September 2022, and the NCSC published the same assessment: Iranian state-linked actors almost certainly carried out the attacks that began on 15 July 2022 (**Confirmed** [52] [53]). The NCSC joined the September 2022 exposure of IRGC-affiliated ransomware operations, and in March 2026 advised UK organizations with a Middle East presence or supply chains to prepare for Iran-linked hacktivist spillover, including DDoS, phishing and ICS targeting (**Confirmed** [55] [56]). The EU's High Representative expressed solidarity with Albania, together with NATO, without naming Iran (**Confirmed** [54]). Albania cut diplomatic relations with Iran on 7 September 2022 (**Reported** [94]). Israel's Foreign Ministry records Iranian operations against Israeli hospitals, water, energy and IT, and the INCD's 2023 annual summary records state attack groups treating hospitals as a legitimate target; Israel is a party to the conflict, which a reader should weigh (**Confirmed** [57] [58]).

Part 8: the defender view

The front doors

For an exposure program, the joint record points to two front doors and a third that is newer (**Inferred** from [15] to [29]):

1. **Known edge-device vulnerabilities.** Exploit Public-Facing Application (`T1190`) and External Remote Services (`T1133`) against VPNs, firewalls, mail servers and remote-access gateways listed in KEV.
2. **Weak or reused credentials.** Password Spraying (`T1110.003`), Multi-Factor Authentication Request Generation (`T1621`), Device Registration (`T1098.005`) and Default Accounts (`T1078.001`).
3. **Internet-facing controllers.** Internet Accessible Device (`T0883`) and Default Credentials (`T1694.001`) in ATT&CK for ICS, the CyberAv3ngers path.

After access the tradecraft is commodity: PowerShell (`T1059.001`), Scheduled Task (`T1053.005`), LSASS Memory (`T1003.001`), Web Shell (`T1505.003`), Protocol Tunneling (`T1572`) and Remote Access Tools (`T1219`) (**Confirmed** for the mappings [1]; the reading is **Inferred**). Targeted espionage against people adds spearphishing and social engineering (`T1566.001`, `T1566.002`), which seven groups each carry in the ATT&CK record (**Confirmed** [1]).

Hunting and detection leads published in the primaries

- Many failed logins, impossible travel, one IP address used across many accounts, MFA registrations from unexpected devices, and access to `ntds.dit` (AA24-290A) (**Confirmed** [25]).
- New accounts and new domain administrators (Windows events 4738, 4728 and 4670), new scheduled tasks, changes to Defender settings, processes accessing LSASS, RDP type-10 logons from one account to many hosts, and outbound connections to ngrok (AA22-320A) (**Confirmed** [21]).
- Unexpected AnyDesk, MeshCentral, Ligolo or ligolo-ng and ngrok, and PowerShell Web Access enabled on servers (AA24-241A) (**Confirmed** [24]).
- Unexpected outbound SSH, SMB and RDP (AA20-259A) (**Confirmed** [15]).
- Unauthorized BitLocker activation (AA22-257A) (**Confirmed** [19]).
- Remote-access log entries and firmware or configuration changes on OT devices (June 2025 fact sheet) (**Confirmed** [27]).

Hardening guidance in the primaries

- Patch KEV-listed internet-facing systems first, use application control, require MFA for webmail and VPN, limit admin rights and add external-sender banners (AA22-055A) (**Confirmed** [18]).
- Rebuild compromised NetScaler devices, remediate the Active Directory forest if compromised, and use separate admin accounts on dedicated workstations (AA20-259A) (**Confirmed** [15]).
- Turn on Credential Guard, limit NTLM and WDigest, and move to phishing-resistant MFA (AA22-320A) (**Confirmed** [21]).
- Review help-desk password reset procedures, disable departing users' accounts, align with NIST digital identity guidance and disable RC4 for Kerberos (AA24-290A) (**Confirmed** [25]).
- Back up BitLocker recovery keys offline (AA22-257A) (**Confirmed** [19]).

- For OT: replace default passwords, take PLCs off the public internet, change the default port, put devices behind a VPN or gateway with MFA, segment networks, keep tested backups of PLC logic, keep PLCs in run mode, and ask manufacturers to ship without default passwords (AA23-335A and the June 2025 fact sheet) (**Confirmed** [23] [27]).

Vulnerability management

The KEV view is unusually clean for this actor set: 19 of the 20 advisory IT CVEs are in KEV and every one is flagged for known ransomware use (**Confirmed** [33]). That makes KEV-first patching a direct defense against the Iranian edge-device clusters, with two cautions from the advisories themselves. Patching does not evict an actor that already holds stolen credentials or a web shell, so AA24-241A asks for a hunt after the patch, and AA22-320A asks defenders to assume compromise on a Horizon server that was not patched promptly (**Confirmed** [24] [21]). The library's Vulnerability Prioritization Reference covers KEV-driven prioritization.

Healthcare-specific steps

- Inventory every internet-facing OT and building system (HVAC, water treatment, life-safety, building automation, environmental controls) and bring it under the same patch and password rules as IT, as the AHA advises and as the June 2021 children's hospital case shows (**Reported** [82]; **Confirmed** [17]).
- Watch identity for spraying and MFA fatigue against clinical and research staff, whose accounts appear in both the AA24-290A victims and the TA453 and APT42 phishing record (**Confirmed** [25] [79] [64]).
- Treat a supplier outage as a security event. When a medical technology vendor is wiped, hospitals that suspend connections early limit their exposure; plan the disconnect in advance (**Inferred** from [93] [90]).
- Protect device-management consoles: restrict bulk wipe and reset rights, require a second approver, and log those actions outside the console (**Inferred** from [59] [93]).

Where the library's countermeasure data is thin

The advisory guidance maps to D3FEND countermeasures that all have library pages, but the library's D3FEND 1.6.0 mapping leaves gaps against this playbook: Multi-factor Authentication (`D3-MFA`) maps Password Spraying but not MFA Request Generation (`T1621`) or External Remote Services (`T1133`); Software Update, Asset Vulnerability Enumeration and Software Inventory do not map `T1190`; Scheduled Job Analysis does not map `T1053.005`; and PowerShell (`T1059.001`) has no D3FEND mapping at all (**Confirmed** [1]). The countermeasures are still the right ones; the library's crosswalk simply does not draw those edges yet.

Part 9: MITRE mappings

This part binds the verified record to MITRE ATT&CK v20.0, CAPEC, CWE and D3FEND. Every ID was checked against the library's data files at commit `e85b559` or the live ATT&CK pages on 3 November 2026, and the technique tables and Navigator layers were regenerated from the library's data files at commit `0c5b7ca` on 11 November 2026 with identical results; IDs that could not be checked are left out. Revoked IDs that advisories still cite are shown under their v20 successors, as `data/attack/superseded_by.jsonl` records them: DLL Side-Loading `T1574.002` as DLL `T1574.001` (AA22-055A), Disable or Modify Tools `T1562.001` as `T1685` (AA22-055A, AA22-320A, AA24-241A), and Downgrade Attack `T1562.010` as `T1689` (AA24-241A) (**Confirmed** [1]).

Enterprise techniques

The table lists every Enterprise technique that ATT&CK v20.0 links to one of the 19 Iran-attributed Enterprise groups, plus every technique named in the eight joint advisories (AA20-259A, AA21-321A, AA22-055A, AA22-257A, AA22-320A, AA23-335A, AA24-241A, AA24-290A). That is 260 techniques: 237 from the groups, 23 only in the advisories, 96 in both (**Confirmed**, computed from [1] and the advisories). Revoked advisory IDs are counted under their successors, so `T1562.001` and `T1562.010` fold into `T1685` and `T1689`. Advisory technique lists were read from each advisory's own ATT&CK tables [15] [17] [18] [19] [21] [23] [24] [25]. Tactic names are the v20 names from the library's `technique_profiles.jsonl`. "Groups" lists the Iran-attributed groups whose ATT&CK record carries the technique, with procedure text and references on each group's page; "Advisories" gives the advisory and its publication date.

Counting each technique once per tactic it serves, the tally is Stealth 38, Credential Access 33, Persistence 31, Discovery 29, Execution 24, Privilege Escalation 24, Collection 23, Resource Development 23, Command and Control 21, Reconnaissance 15, Initial Access 14, Defense Impairment 13, Lateral Movement 10, Impact 10 and Exfiltration 5 (computed from [1]).

The most-reported techniques, scored as the number of groups plus the number of advisories that name them: Obtain Capabilities: Tool `T1588.002` (14 groups, 4 advisories), PowerShell `T1059.001` (11, 5), Ingress Tool Transfer `T1105` (10, 4), LSASS Memory `T1003.001` (10, 3), Scheduled Task `T1053.005` (8, 5), Exploit Public-Facing Application `T1190` (7, 5) and Archive via Utility `T1560.001` (8, 4). Password Spraying `T1110.003` is mapped to five groups (APT33, Leafminer, Silent Librarian, Agrius, NIMBUS MANTICORE); `T1621` and `T1098.005` appear only in AA24-290A (**Confirmed** [1] [25]).

All 260 Enterprise techniques (expand)

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1001.001	Data Obfuscation: Junk Data	Command and Control	none	AA22-055A, 2022-02-24	[18]
T1003	OS Credential Dumping	Credential Access	G0087	none	[1]
T1003.001	OS Credential Dumping: LSASS Memory	Credential Access	G0003, G0049, G0059, G0064, G0069, G0077, G0087, G0117, G1030, G1055	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA22-320A, 2022-11-16	[1], [15], [18], [21]
T1003.002	OS Credential Dumping: Security Account Manager	Credential Access	G1030	none	[1]
T1003.003	OS Credential Dumping: NTDS	Credential Access	G0117	AA20-259A, 2020-09-15	[1], [15]
T1003.004	OS Credential Dumping: LSA Secrets	Credential Access	G0049, G0064, G0069, G0077	AA22-055A, 2022-02-24	[1], [18]
T1003.005	OS Credential Dumping: Cached Domain Credentials	Credential Access	G0049, G0064, G0069, G0077	AA22-055A, 2022-02-24	[1], [18]
T1003.006	OS Credential Dumping: DCSync	Credential Access	G1059	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1005	Data from Local System	Collection	G0049, G0059, G0087, G0117, G1012, G1030, G1055	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA24-290A, 2024-10-16	[1], [15], [18], [25]
T1007	System Service Discovery	Discovery	G0049, G1059	none	[1]
T1008	Fallback Channels	Command and Control	G0049	none	[1]
T1012	Query Registry	Discovery	G0049, G0087, G0117	AA24-241A, 2024-08-28	[1], [24]
T1016	System Network Configuration Discovery	Discovery	G0049, G0059, G0069, G1009, G1044	AA22-055A, 2022-02-24	[1], [18]
T1016.001	System Network Configuration Discovery: Internet Connection Discovery	Discovery	G0059	AA22-320A, 2022-11-16	[1], [21]
T1016.002	System Network Configuration Discovery: Wi-Fi Discovery	Discovery	G0059	none	[1]
T1018	Remote System Discovery	Discovery	G0059, G0077, G0087, G0117, G1030	AA20-259A, 2020-09-15; AA22-320A, 2022-11-16; AA24-290A, 2024-10-16	[1], [15], [21], [25]
T1021	Remote Services	Lateral Movement	none	AA20-259A, 2020-09-15	[15]
T1021.001	Remote Services: Remote Desktop Protocol	Lateral Movement	G0049, G0059, G0087, G0117, G1030, G1055, G1059	AA20-259A, 2020-09-15; AA22-320A, 2022-11-16; AA24-290A, 2024-10-16	[1], [15], [21], [25]
T1021.002	Remote Services: SMB/Windows Admin Shares	Lateral Movement	G0087, G0117, G1009, G1059	AA20-259A, 2020-09-15	[1], [15]
T1021.004	Remote Services: SSH	Lateral Movement	G0049, G0087, G0117	AA20-259A, 2020-09-15	[1], [15]
T1021.005	Remote Services: VNC	Lateral Movement	G0117	AA20-259A, 2020-09-15	[1], [15]
T1025	Data from Removable Media	Collection	G0049	none	[1]
T1027	Obfuscated Files or Information	Stealth	G1059	AA22-055A, 2022-02-24	[1], [18]
T1027.002	Obfuscated Files or Information: Software Packing	Stealth	G0087	AA20-259A, 2020-09-15	[1], [15]
T1027.003	Obfuscated Files or Information: Steganography	Stealth	G0069	AA22-055A, 2022-02-24	[1], [18]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1027.004	Obfuscated Files or Information: Compile After Delivery	Stealth	G0069	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24	[1], [15], [18]
T1027.005	Obfuscated Files or Information: Indicator Removal from Tools	Stealth	G0049	none	[1]
T1027.010	Obfuscated Files or Information: Command Obfuscation	Stealth	G0059, G0069, G0077, G0117	none	[1]
T1027.013	Obfuscated Files or Information: Encrypted/Encoded File	Stealth	G0043, G0049, G0059, G0064, G0087, G0117, G1009	none	[1]
T1027.015	Obfuscated Files or Information: Compression	Stealth	G1055, G1059	none	[1]
T1033	System Owner/User Discovery	Discovery	G0049, G0059, G0069, G0087	AA22-055A, 2022-02-24	[1], [18]
T1036	Masquerading	Stealth	G0049, G1030	none	[1]
T1036.004	Masquerading: Masquerade Task or Service	Stealth	G0059, G0117, G1055, G1059	AA20-259A, 2020-09-15	[1], [15]
T1036.005	Masquerading: Match Legitimate Resource Name or Location	Stealth	G0049, G0059, G0069, G0087, G0117, G1044, G1055, G1059	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24	[1], [15], [18]
T1036.010	Masquerading: Masquerade Account Name	Stealth	G0059	none	[1]
T1039	Data from Network Shared Drive	Collection	G0117, G1059	AA20-259A, 2020-09-15	[1], [15]
T1040	Network Sniffing	Credential Access, Discovery	G0064	none	[1]
T1041	Exfiltration Over C2 Channel	Exfiltration	G0069, G0087, G1012, G1030, G1055	AA22-055A, 2022-02-24	[1], [18]
T1046	Network Service Discovery	Discovery	G0049, G0059, G0077, G0087, G0117, G1030, G1059	none	[1]
T1047	Windows Management Instrumentation	Execution	G0049, G0059, G0069, G1044, G1055, G1059	AA22-055A, 2022-02-24	[1], [18]
T1048.002	Exfiltration Over Alternative Protocol: Exfiltration Over Asymmetric Encrypted Non-C2 Protocol	Exfiltration	G1012	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1048.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol	Exfiltration	G0049, G0064	none	[1]
T1049	System Network Connections Discovery	Discovery	G0049, G0059, G0069, G1059	AA22-055A, 2022-02-24	[1], [18]
T1053	Scheduled Task/Job	Execution, Persistence, Privilege Escalation	none	AA24-241A, 2024-08-28	[24]
T1053.003	Scheduled Task/Job: Cron	Execution, Persistence, Privilege Escalation	none	AA20-259A, 2020-09-15	[15]
T1053.005	Scheduled Task/Job: Scheduled Task	Execution, Persistence, Privilege Escalation	G0049, G0059, G0064, G0069, G0087, G0117, G1044, G1059	AA20-259A, 2020-09-15; AA21-321A, 2021-11-17; AA22-055A, 2022-02-24; AA22-257A, 2022-09-14; AA22-320A, 2022-11-16	[1], [15], [17], [18], [19], [21]
T1055.013	Process Injection: Process Doppelgänging	Stealth, Privilege Escalation	G0077	none	[1]
T1056	Input Capture	Collection, Credential Access	G0087, G1044	AA24-241A, 2024-08-28	[1], [24]
T1056.001	Input Capture: Keylogging	Collection, Credential Access	G0043, G0049, G0059, G0087, G0130, G1044	none	[1]
T1057	Process Discovery	Discovery	G0049, G0059, G0069, G1059	AA22-055A, 2022-02-24	[1], [18]
T1059	Command and Scripting Interpreter	Execution	G0049, G0087, G0117	none	[1]
T1059.001	Command and Scripting Interpreter: PowerShell	Execution	G0049, G0052, G0059, G0064, G0069, G0087, G0117, G1012, G1044, G1055, G1059	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA22-320A, 2022-11-16; AA24-241A, 2024-08-28; AA24-290A, 2024-10-16	[1], [15], [18], [21], [24], [25]
T1059.003	Command and Scripting Interpreter: Windows Command Shell	Execution	G0049, G0059, G0069, G0117, G1030	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24	[1], [15], [18]
T1059.005	Command and Scripting Interpreter: Visual Basic	Execution	G0049, G0059, G0064, G0069, G0087, G1044	AA22-055A, 2022-02-24	[1], [18]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1059.006	Command and Scripting Interpreter: Python	Execution	G0069, G0087, G1055	AA22-055A, 2022-02-24	[1], [18]
T1059.007	Command and Scripting Interpreter: JavaScript	Execution	G0069, G0077, G1059	AA22-055A, 2022-02-24	[1], [18]
T1059.010	Command and Scripting Interpreter: AutoHotKey & AutoIT	Execution	G0087	none	[1]
T1068	Exploitation for Privilege Escalation	Privilege Escalation	G0049, G0064	AA24-290A, 2024-10-16	[1], [25]
T1069.001	Permission Groups Discovery: Local Groups	Discovery	G0049	none	[1]
T1069.002	Permission Groups Discovery: Domain Groups	Discovery	G0049	AA24-290A, 2024-10-16	[1], [25]
T1069.003	Permission Groups Discovery: Cloud Groups	Discovery	none	AA24-290A, 2024-10-16	[25]
T1070	Indicator Removal	Stealth	G1044	none	[1]
T1070.003	Indicator Removal: Clear Command History	Stealth	G0059	none	[1]
T1070.004	Indicator Removal: File Deletion	Stealth	G0043, G0049, G0059, G0087, G1059	AA20-259A, 2020-09-15; AA22-320A, 2022-11-16	[1], [15], [21]
T1070.007	Indicator Removal: Clear Network Connection History and Configurations	Stealth	G1059	none	[1]
T1070.008	Indicator Removal: Clear Mailbox Data	Stealth	G1044	none	[1]
T1071	Application Layer Protocol	Command and Control	G0059, G1059	none	[1]
T1071.001	Application Layer Protocol: Web Protocols	Command and Control	G0049, G0059, G0064, G0069, G0087, G1044, G1055	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA24-290A, 2024-10-16	[1], [15], [18], [25]
T1071.004	Application Layer Protocol: DNS	Command and Control	G0049, G0087	none	[1]
T1072	Software Deployment Tools	Execution, Lateral Movement	G1055, G1059	none	[1]
T1074	Data Staged	Collection	G1055	none	[1]
T1074.001	Data Staged: Local Data Staging	Collection	G0069, G0087, G1030	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1078	Valid Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	G0049, G0064, G0087, G0117, G0122, G1005, G1055	AA24-290A, 2024-10-16	[1], [25]
T1078.001	Valid Accounts: Default Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	G0059	AA22-320A, 2022-11-16; AA23-335A, 2023-12-01	[1], [21], [23]
T1078.002	Valid Accounts: Domain Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	G0049, G0059, G1030, G1055, G1059	AA24-241A, 2024-08-28	[1], [24]
T1078.003	Valid Accounts: Local Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	G1059	AA24-241A, 2024-08-28	[1], [24]
T1078.004	Valid Accounts: Cloud Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	G0064, G1055, G1059	AA24-290A, 2024-10-16	[1], [25]
T1082	System Information Discovery	Discovery	G0049, G0059, G0069, G1009, G1012, G1044, G1055	AA22-055A, 2022-02-24; AA24-290A, 2024-10-16	[1], [18], [25]
T1083	File and Directory Discovery	Discovery	G0059, G0069, G0077, G0087, G0117	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24	[1], [15], [18]
T1087	Account Discovery	Discovery	none	AA20-259A, 2020-09-15	[15]
T1087.001	Account Discovery: Local Account	Discovery	G0049, G0117, G1009, G1044, G1059	none	[1]
T1087.002	Account Discovery: Domain Account	Discovery	G0049, G0069, G0117, G1055, G1059	AA22-055A, 2022-02-24; AA24-290A, 2024-10-16	[1], [18], [25]
T1087.003	Account Discovery: Email Account	Discovery	G0059	none	[1]
T1090	Proxy	Command and Control	G0052, G0059, G0069, G0117, G1005	AA22-320A, 2022-11-16	[1], [21]
T1090.001	Proxy: Internal Proxy	Command and Control	G0087	none	[1]
T1090.002	Proxy: External Proxy	Command and Control	G0069, G0087	AA22-055A, 2022-02-24	[1], [18]
T1098	Account Manipulation	Persistence, Privilege Escalation	G1055, G1059	AA22-320A, 2022-11-16; AA24-241A, 2024-08-28	[1], [21], [24]
T1098.002	Account Manipulation: Additional Email Delegate Permissions	Persistence, Privilege Escalation	G0059	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1098.005	Account Manipulation: Device Registration	Persistence, Privilege Escalation	none	AA24-290A, 2024-10-16	[25]
T1098.007	Account Manipulation: Additional Local or Domain Groups	Persistence, Privilege Escalation	G0059	none	[1]
T1102	Web Service	Command and Control	G0117, G1044, G1055	none	[1]
T1102.002	Web Service: Bidirectional Communication	Command and Control	G0059, G0069, G0087, G1005	AA22-055A, 2022-02-24	[1], [18]
T1104	Multi-Stage Channels	Command and Control	G0069	AA22-055A, 2022-02-24	[1], [18]
T1105	Ingress Tool Transfer	Command and Control	G0049, G0059, G0064, G0069, G0087, G0117, G0130, G1009, G1055, G1059	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA22-320A, 2022-11-16; AA24-290A, 2024-10-16	[1], [15], [18], [21], [25]
T1106	Native API	Execution	G1059	none	[1]
T1110	Brute Force	Credential Access	G0049, G0087, G0117, G1030, G1055	AA23-335A, 2023-12-01; AA24-290A, 2024-10-16	[1], [23], [25]
T1110.001	Brute Force: Password Guessing	Credential Access	G1055	none	[1]
T1110.003	Brute Force: Password Spraying	Credential Access	G0064, G0077, G0122, G1030, G1059	AA24-290A, 2024-10-16	[1], [25]
T1110.004	Brute Force: Credential Stuffing	Credential Access	G1055	none	[1]
T1111	Multi-Factor Authentication Interception	Credential Access	G1044	none	[1]
T1112	Modify Registry	Defense Impairment, Persistence	G0049, G0059, G1044	none	[1]
T1113	Screen Capture	Collection	G0043, G0049, G0059, G0069, G0087, G1044, G1055	AA22-055A, 2022-02-24	[1], [18]
T1114	Email Collection	Collection	G0059, G0122	none	[1]
T1114.001	Email Collection: Local Email Collection	Collection	G0059, G1059	none	[1]
T1114.002	Email Collection: Remote Email Collection	Collection	G0059, G0077, G1055	none	[1]
T1114.003	Email Collection: Email Forwarding Rule	Collection	G0122	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1115	Clipboard Data	Collection	G0049, G0087	none	[1]
T1119	Automated Collection	Collection	G0049, G1030, G1055	none	[1]
T1120	Peripheral Device Discovery	Discovery	G0049	none	[1]
T1123	Audio Capture	Collection	G1055	none	[1]
T1124	System Time Discovery	Discovery	G1012	none	[1]
T1125	Video Capture	Collection	G1055	none	[1]
T1132.001	Data Encoding: Standard Encoding	Command and Control	G0064, G0069, G1044	AA22-055A, 2022-02-24	[1], [18]
T1132.002	Data Encoding: Non-Standard Encoding	Command and Control	none	AA22-055A, 2022-02-24	[18]
T1133	External Remote Services	Persistence, Initial Access	G0049, G1055	AA24-241A, 2024-08-28; AA24-290A, 2024-10-16	[1], [24], [25]
T1135	Network Share Discovery	Discovery	G0087	none	[1]
T1136.001	Create Account: Local Account	Persistence	G0059, G0077, G0087, G0117, G1059	AA21-321A, 2021-11-17; AA22-257A, 2022-09-14; AA22-320A, 2022-11-16; AA24-241A, 2024-08-28	[1], [17], [19], [21], [24]
T1136.002	Create Account: Domain Account	Persistence	none	AA21-321A, 2021-11-17; AA22-257A, 2022-09-14; AA22-320A, 2022-11-16	[17], [19], [21]
T1137.001	Office Application Startup: Office Template Macros	Persistence	G0069	AA22-055A, 2022-02-24	[1], [18]
T1137.004	Office Application Startup: Outlook Home Page	Persistence	G0049	none	[1]
T1140	Deobfuscate/Decode Files or Information	Stealth	G0049, G0069, G0087, G1030	AA22-055A, 2022-02-24	[1], [18]
T1189	Drive-by Compromise	Initial Access	G0059, G0077, G1012	none	[1]
T1190	Exploit Public-Facing Application	Initial Access	G0059, G0069, G0087, G0117, G1009, G1030, G1055	AA20-259A, 2020-09-15; AA21-321A, 2021-11-17; AA22-257A, 2022-09-14; AA22-320A, 2022-11-16; AA24-241A, 2024-08-28	[1], [15], [17], [19], [21], [24]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1195	Supply Chain Compromise	Initial Access	G0049	none	[1]
T1197	BITS Jobs	Stealth, Persistence, Execution	G0087	none	[1]
T1199	Trusted Relationship	Initial Access	G1005, G1055, G1059	none	[1]
T1201	Password Policy Discovery	Discovery	G0049	none	[1]
T1202	Indirect Command Execution	Stealth	none	AA24-290A, 2024-10-16	[25]
T1203	Exploitation for Client Execution	Execution	G0049, G0064, G0069	AA22-055A, 2022-02-24	[1], [18]
T1204.001	User Execution: Malicious Link	Execution	G0049, G0059, G0064, G0069, G0087, G1059	AA22-055A, 2022-02-24	[1], [18]
T1204.002	User Execution: Malicious File	Execution	G0049, G0059, G0064, G0069, G0087, G0130, G1012, G1055, G1059	AA22-055A, 2022-02-24	[1], [18]
T1204.004	User Execution: Malicious Copy and Paste	Execution	G0069	none	[1]
T1210	Exploitation of Remote Services	Lateral Movement	G0069, G0117	none	[1]
T1213	Data from Information Repositories	Collection	G1059	AA20-259A, 2020-09-15	[1], [15]
T1213.002	Data from Information Repositories: Sharepoint	Collection	G1055, G1059	none	[1]
T1213.005	Data from Information Repositories: Messaging Applications	Collection	G0117, G1059	none	[1]
T1217	Browser Information Discovery	Discovery	G0117	AA20-259A, 2020-09-15	[1], [15]
T1218.001	System Binary Proxy Execution: Compiled HTML File	Stealth	G0049	none	[1]
T1218.003	System Binary Proxy Execution: CMSTP	Stealth	G0069	AA22-055A, 2022-02-24	[1], [18]
T1218.005	System Binary Proxy Execution: Mshta	Stealth	G0069	AA22-055A, 2022-02-24	[1], [18]
T1218.011	System Binary Proxy Execution: Rundll32	Stealth	G0052, G0059, G0069	AA22-055A, 2022-02-24	[1], [18]
T1219	Remote Access Tools	Command and Control	G0049	AA22-055A, 2022-02-24; AA24-241A, 2024-08-28	[1], [18], [24]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1219.002	Remote Access Tools: Remote Desktop Software	Command and Control	G0069, G1055	none	[1]
T1480	Execution Guardrails	Stealth	none	AA22-055A, 2022-02-24	[18]
T1482	Domain Trust Discovery	Discovery	G0059	AA24-241A, 2024-08-28; AA24-290A, 2024-10-16	[1], [24], [25]
T1484.001	Domain or Tenant Policy Modification: Group Policy Modification	Defense Impairment, Privilege Escalation	G1055	none	[1]
T1484.002	Domain or Tenant Policy Modification: Trust Modification	Defense Impairment, Privilege Escalation	none	AA24-290A, 2024-10-16	[25]
T1485	Data Destruction	Impact	G1055	none	[1]
T1486	Data Encrypted for Impact	Impact	G0059, G1055	AA21-321A, 2021-11-17; AA22-257A, 2022-09-14	[1], [17], [19]
T1490	Inhibit System Recovery	Impact	G1055	none	[1]
T1491.001	Defacement: Internal Defacement	Impact	none	AA23-335A, 2023-12-01	[23]
T1497.001	Virtualization/Sandbox Evasion: System Checks	Stealth, Discovery	G0049	none	[1]
T1499	Endpoint Denial of Service	Impact	none	AA23-335A, 2023-12-01	[23]
T1505	Server Software Component	Persistence	none	AA24-241A, 2024-08-28	[24]
T1505.003	Server Software Component: Web Shell	Persistence	G0049, G0059, G0087, G0117, G1009, G1012, G1030	AA20-259A, 2020-09-15; AA24-241A, 2024-08-28	[1], [15], [24]
T1518	Software Discovery	Discovery	G0069	AA22-055A, 2022-02-24	[1], [18]
T1518.001	Software Discovery: Security Software Discovery	Discovery	G0069, G1044	AA22-055A, 2022-02-24	[1], [18]
T1530	Data from Cloud Storage	Collection	G0117, G1044	AA20-259A, 2020-09-15	[1], [15]
T1531	Account Access Removal	Impact	none	AA23-335A, 2023-12-01	[23]
T1534	Internal Spearphishing	Lateral Movement	G0069, G1059	none	[1]
T1539	Steal Web Session Cookie	Credential Access	G1044	none	[1]
T1543.003	Create or Modify System Process: Windows Service	Persistence, Privilege Escalation	G0049, G1030	none	[1]
T1546.003	Event Triggered Execution: Windows Management Instrumentation Event Subscription	Privilege Escalation, Persistence	G0064	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1546.008	Event Triggered Execution: Accessibility Features	Privilege Escalation, Persistence	G0117	AA20-259A, 2020-09-15	[1], [15]
T1546.010	Event Triggered Execution: AppInit DLLs	Privilege Escalation, Persistence	G0087	none	[1]
T1547	Boot or Logon Autostart Execution	Persistence, Privilege Escalation	G1044	none	[1]
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	Persistence, Privilege Escalation	G0059, G0064, G0069, G0087, G1055, G1059	AA22-055A, 2022-02-24	[1], [18]
T1547.009	Boot or Logon Autostart Execution: Shortcut Modification	Persistence, Privilege Escalation	G0087	none	[1]
T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	Privilege Escalation	G0069	AA22-055A, 2022-02-24	[1], [18]
T1552	Unsecured Credentials	Credential Access	G1059	none	[1]
T1552.001	Unsecured Credentials: Credentials In Files	Credential Access	G0049, G0064, G0069, G0077, G0117	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24	[1], [15], [18]
T1552.002	Unsecured Credentials: Credentials in Registry	Credential Access	G1055	none	[1]
T1552.006	Unsecured Credentials: Group Policy Preferences	Credential Access	G0064	none	[1]
T1553.002	Subvert Trust Controls: Code Signing	Defense Impairment	G0049, G0052, G1009	none	[1]
T1553.006	Subvert Trust Controls: Code Signing Policy Modification	Defense Impairment	G0087	none	[1]
T1555	Credentials from Password Stores	Credential Access	G0049, G0064, G0069, G0077, G0087	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA22-320A, 2022-11-16; AA24-290A, 2024-10-16	[1], [15], [18], [21], [25]
T1555.003	Credentials from Password Stores: Credentials from Web Browsers	Credential Access	G0049, G0064, G0069, G0077, G0130, G1044	AA22-055A, 2022-02-24	[1], [18]
T1555.004	Credentials from Password Stores: Windows Credential Manager	Credential Access	G0049	none	[1]
T1555.005	Credentials from Password Stores: Password Managers	Credential Access	G0117	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1556	Modify Authentication Process	Defense Impairment, Persistence, Credential Access	none	AA24-290A, 2024-10-16	[25]
T1556.002	Modify Authentication Process: Password Filter DLL	Defense Impairment, Persistence, Credential Access	G0049	none	[1]
T1556.006	Modify Authentication Process: Multi-Factor Authentication	Defense Impairment, Persistence, Credential Access	none	AA24-290A, 2024-10-16	[25]
T1557.002	Adversary-in-the-Middle: ARP Cache Poisoning	Credential Access, Collection	G0003	none	[1]
T1558	Steal or Forge Kerberos Tickets	Credential Access	none	AA20-259A, 2020-09-15	[15]
T1558.003	Steal or Forge Kerberos Tickets: Kerberoasting	Credential Access	G1059	AA24-290A, 2024-10-16	[1], [25]
T1559.001	Inter-Process Communication: Component Object Model	Execution	G0069	AA22-055A, 2022-02-24	[1], [18]
T1559.002	Inter-Process Communication: Dynamic Data Exchange	Execution	G0069	AA22-055A, 2022-02-24	[1], [18]
T1560.001	Archive Collected Data: Archive via Utility	Collection	G0052, G0059, G0064, G0069, G0087, G0117, G1030, G1055	AA20-259A, 2020-09-15; AA21-321A, 2021-11-17; AA22-055A, 2022-02-24; AA22-257A, 2022-09-14	[1], [15], [17], [18], [19]
T1560.003	Archive Collected Data: Archive via Custom Method	Collection	G0052	none	[1]
T1561.001	Disk Wipe: Disk Content Wipe	Impact	G1055	none	[1]
T1561.002	Disk Wipe: Disk Structure Wipe	Impact	G1055	none	[1]
T1563.002	Remote Service Session Hijacking: RDP Hijacking	Lateral Movement	G1059	AA20-259A, 2020-09-15	[1], [15]
T1564.003	Hide Artifacts: Hidden Window	Stealth	G0052, G0059, G1055	none	[1]
T1565.001	Data Manipulation: Stored Data Manipulation	Impact	none	AA23-335A, 2023-12-01	[23]
T1566	Phishing	Initial Access	G0069, G1055	none	[1]
T1566.001	Phishing: Spearphishing Attachment	Initial Access	G0049, G0064, G0069, G0087, G0130, G1012, G1059	AA22-055A, 2022-02-24	[1], [18]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1566.002	Phishing: Spearphishing Link	Initial Access	G0049, G0059, G0064, G0069, G0087, G1044, G1059	AA22-055A, 2022-02-24	[1], [18]
T1566.003	Phishing: Spearphishing via Service	Initial Access	G0049, G0059, G0130, G1012	none	[1]
T1567	Exfiltration Over Web Service	Exfiltration	G0059	none	[1]
T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage	Exfiltration	G0069, G1005	none	[1]
T1569.002	System Services: Service Execution	Execution	G0087	none	[1]
T1570	Lateral Tool Transfer	Lateral Movement	G0059, G1030	none	[1]
T1571	Non-Standard Port	Command and Control	G0059, G0064, G0069	none	[1]
T1572	Protocol Tunneling	Command and Control	G0049, G0059, G0117, G1055	AA20-259A, 2020-09-15; AA22-055A, 2022-02-24; AA24-241A, 2024-08-28; AA24-290A, 2024-10-16	[1], [15], [18], [24], [25]
T1573	Encrypted Channel	Command and Control	G0059	none	[1]
T1573.001	Encrypted Channel: Symmetric Cryptography	Command and Control	G0064, G0069	none	[1]
T1573.002	Encrypted Channel: Asymmetric Cryptography	Command and Control	G0049, G1044	none	[1]
T1574.001	Hijack Execution Flow: DLL	Stealth, Execution	G0069, G1059	AA22-055A, 2022-02-24	[1], [18]
T1574.014	Hijack Execution Flow: AppDomainManager	Stealth, Execution	G1059	none	[1]
T1583	Acquire Infrastructure	Resource Development	G1030	none	[1]
T1583.001	Acquire Infrastructure: Domains	Resource Development	G0049, G0059, G0069, G0122, G1012, G1044, G1055, G1059	none	[1]
T1583.003	Acquire Infrastructure: Virtual Private Server	Resource Development	G1012, G1044, G1055, G1059	none	[1]
T1583.004	Acquire Infrastructure: Server	Resource Development	G1012, G1055	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1583.006	Acquire Infrastructure: Web Services	Resource Development	G0059, G0069, G1005, G1055, G1059	AA22-055A, 2022-02-24	[1], [18]
T1584.001	Compromise Infrastructure: Domains	Resource Development	G0059	none	[1]
T1584.006	Compromise Infrastructure: Web Services	Resource Development	G1012	none	[1]
T1585	Establish Accounts	Resource Development	G0117	none	[1]
T1585.001	Establish Accounts: Social Media Accounts	Resource Development	G0003, G0059, G0117, G1012, G1055, G1059	none	[1]
T1585.002	Establish Accounts: Email Accounts	Resource Development	G0059, G0122, G1012, G1044, G1055	none	[1]
T1586.002	Compromise Accounts: Email Accounts	Resource Development	G0049, G0059	none	[1]
T1587.001	Develop Capabilities: Malware	Resource Development	G0003, G0049, G1009, G1055, G1059	none	[1]
T1588.001	Obtain Capabilities: Malware	Resource Development	G0069, G1055	AA21-321A, 2021-11-17; AA22-257A, 2022-09-14	[1], [17], [19]
T1588.002	Obtain Capabilities: Tool	Resource Development	G0003, G0049, G0052, G0059, G0064, G0069, G0077, G0087, G0122, G1005, G1009, G1044, G1055, G1059	AA21-321A, 2021-11-17; AA22-055A, 2022-02-24; AA22-257A, 2022-09-14; AA24-290A, 2024-10-16	[1], [17], [18], [19], [25]
T1588.003	Obtain Capabilities: Code Signing Certificates	Resource Development	G0049, G1059	none	[1]
T1588.004	Obtain Capabilities: Digital Certificates	Resource Development	G0122	none	[1]
T1588.007	Obtain Capabilities: Artificial Intelligence	Resource Development	G1059	none	[1]
T1589	Gather Victim Identity Information	Reconnaissance	G0059, G1055	AA24-290A, 2024-10-16	[1], [25]
T1589.001	Gather Victim Identity Information: Credentials	Reconnaissance	G0059	none	[1]
T1589.002	Gather Victim Identity Information: Email Addresses	Reconnaissance	G0059, G0122	AA22-055A, 2022-02-24	[1], [18]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1589.003	Gather Victim Identity Information: Employee Names	Reconnaissance	G0122	none	[1]
T1590.004	Gather Victim Network Information: Network Topology	Reconnaissance	G0069	none	[1]
T1590.005	Gather Victim Network Information: IP Addresses	Reconnaissance	G0059	none	[1]
T1591	Gather Victim Org Information	Reconnaissance	G1059	none	[1]
T1591.001	Gather Victim Org Information: Determine Physical Locations	Reconnaissance	G0059	none	[1]
T1592.002	Gather Victim Host Information: Software	Reconnaissance	G0059	none	[1]
T1594	Search Victim-Owned Websites	Reconnaissance	G0122	none	[1]
T1595.002	Active Scanning: Vulnerability Scanning	Reconnaissance	G0059, G1055	none	[1]
T1596	Search Open Technical Databases	Reconnaissance	none	AA24-241A, 2024-08-28	[24]
T1598.001	Phishing for Information: Spearphishing Service	Reconnaissance	G1059	none	[1]
T1598.003	Phishing for Information: Spearphishing Link	Reconnaissance	G0059, G0122, G1012	none	[1]
T1608.001	Stage Capabilities: Upload Malware	Resource Development	G0049, G1044, G1059	none	[1]
T1608.004	Stage Capabilities: Drive-by Target	Resource Development	G1012	none	[1]
T1608.005	Stage Capabilities: Link Target	Resource Development	G0122, G1059	none	[1]
T1608.006	Stage Capabilities: SEO Poisoning	Resource Development	G1059	none	[1]
T1621	Multi-Factor Authentication Request Generation	Credential Access	none	AA24-290A, 2024-10-16	[25]
T1649	Steal or Forge Authentication Certificates	Credential Access	G1059	none	[1]
T1651	Cloud Administration Command	Execution	G1055	none	[1]
T1657	Financial Theft	Impact	G1055	AA24-241A, 2024-08-28	[1], [24]
T1678	Delay Execution	Stealth	G1059	none	[1]

ID	Technique	Tactic (v20)	Groups	Advisories	Sources
T1679	Selective Exclusion	Stealth	G1055	none	[1]
T1682	Query Public AI Services	Reconnaissance	G1044	none	[1]
T1683.001	Generate Content: Written Content	Resource Development	G1059	none	[1]
T1683.002	Generate Content: Audio-Visual Content	Resource Development	G1059	none	[1]
T1684.001	Social Engineering: Impersonation	Stealth	G0069, G1044, G1055, G1059	none	[1]
T1685	Disable or Modify Tools	Defense Impairment	G0059, G0069, G1030	AA22-055A, 2022-02-24; AA22-320A, 2022-11-16; AA24-241A, 2024-08-28	[1], [18], [21], [24]
T1685.001	Disable or Modify Tools: Disable or Modify Windows Event Log	Defense Impairment	G0059	none	[1]
T1685.003	Disable or Modify Tools: Modify or Spoof Tool UI	Defense Impairment	G1059	none	[1]
T1686.003	Disable or Modify System Firewall: Windows Host Firewall	Defense Impairment	G0049, G0059, G1009, G1055	none	[1]
T1689	Downgrade Attack	Defense Impairment	none	AA24-241A, 2024-08-28	[24]
T1697	Steal Application Specific Passwords	Credential Access	G1044	none	[1]

Vendor primaries add technique statements that are already in the table: Microsoft maps Peach Sandstorm's password spraying to `T1110.003` and its exploitation of CVE-2022-47966 and CVE-2022-26134 to `T1190` (**Confirmed** [71]).

ATT&CK for ICS

ATT&CK for ICS links 24 techniques to three Iran-attributed groups: 17 to CyberAv3ngers, 6 to OilRig and 3 to APT33 (**Confirmed** [1]). The library has no ICS technique pages under `mitre/`, so the IDs link to attack.mitre.org; see also the library's ICS ATT&CK Atlas. The live CyberAv3ngers page lists 19 technique rows; the two not in the library's mapping are Denial of Service `T0814` and Loss of Productivity and Revenue `T0828` (**Confirmed** [12] [1]). AA23-335A itself lists only Enterprise IDs for the Unitronics activity (`T1110`, `T1078.001`, `T1565.001`, `T1531`, `T1499`, `T1491.001`), which are in the Enterprise table (**Confirmed** [23]).

ID	Technique	Tactic	Groups	Sources
T0817	Drive-by Compromise	Initial Access	G0049 OilRig	[1]
T0826	Loss of Availability	Impact	G1027 CyberAv3ngers	[1], [12]
T0827	Loss of Control	Impact	G1027 CyberAv3ngers	[1], [12]
T0829	Loss of View	Impact	G1027 CyberAv3ngers	[1], [12]

ID	Technique	Tactic	Groups	Sources
T0838	Modify Alarm Settings	Inhibit Response Function	G1027 CyberAv3ngers	[1], [12]
T0843	Program Download	Lateral Movement	G1027 CyberAv3ngers	[1], [12]
T0845	Program Upload	Collection	G1027 CyberAv3ngers	[1], [12]
T0852	Screen Capture	Collection	G0064 APT33	[1]
T0859	Valid Accounts	Persistence, Lateral Movement	G0049 OilRig	[1]
T0865	Spearphishing Attachment	Initial Access	G0049 OilRig, G0064 APT33	[1]
T0869	Standard Application Layer Protocol	Command and Control	G0049 OilRig	[1]
T0871	Execution through API	Execution	G1027 CyberAv3ngers	[1], [12]
T0879	Damage to Property	Impact	G1027 CyberAv3ngers	[1], [12]
T0880	Loss of Safety	Impact	G1027 CyberAv3ngers	[1], [12]
T0883	Internet Accessible Device	Initial Access	G1027 CyberAv3ngers	[1], [12]
T0885	Commonly Used Port	Command and Control	G1027 CyberAv3ngers	[1], [12]
T0886	Remote Services	Initial Access, Lateral Movement	G1027 CyberAv3ngers	[1], [12]
T0889	Modify Program	Persistence	G1027 CyberAv3ngers	[1], [12]
T0892	Change Credential	Inhibit Response Function	G1027 CyberAv3ngers	[1], [12]
T1692.001	Unauthorized Message: Command Message	Evasion, Impair Process Control	G1027 CyberAv3ngers	[1], [12]
T1694.001	Insecure Credentials: Default Credentials	Persistence, Lateral Movement	G1027 CyberAv3ngers	[1], [12]
T1695	Block Communications	Inhibit Response Function	G1027 CyberAv3ngers	[1], [12]
T1701.001	Command and Scripting Interpreter: PowerShell	Execution	G0049 OilRig, G0064 APT33	[1]
T1701.004	Command and Scripting Interpreter: Visual Basic	Execution	G0049 OilRig	[1]

ATT&CK for Mobile

UNC788 (G1029) has one Mobile technique, Phishing T1660 (Initial Access), and one software entry (**Confirmed** [1]). The library generates no Mobile group page.

Groups, campaigns and software

Object	ID	Relationship in ATT&CK v20.0	Source
The 19 Enterprise groups	see Part 1	Iran-attributed in their own description	[1]
CyberAv3ngers	G1027	ICS-only group; no library page	[1] [12]
UNC788	G1029	Mobile-only group; no library page	[1]
UNC3890 campaign	C0010	2020-12 to 2022-08; Israeli shipping, government, aviation, energy and healthcare; no group object	[1]
HomeLand Justice	C0038	2021-05 to 2022-09; Albania; the Groups table lists only VOID MANTICORE, and HEXANE appears in the narrative only	[13]
Outer Space	C0042	OilRig, 2021	[1]
Juicy Mix	C0044	OilRig, 2022	[1]
HEXANE	G1001	Not Iran-attributed in its own description; named as Iran-nexus in C0038	[1] [13]
Shamoon	S0140	First used in 2012 by a group calling itself Cutting Sword of Justice	[1]
ZeroCleare	S1151	RawDisk-based wiper since at least 2019; linked to OilRig; used in C0038	[1] [13]
CHIMNEYSWEEP, ROADSWEEP	S1149, S1150	Used in C0038	[13]
IOControl	S9046	Attributed to IRGC-affiliated CyberAv3ngers	[1]
Havij	S0224	Iranian ITSecTeam tool used by Ajax Security Team	[1]
RDAT	S0495	OilRig	[1]
Group software	111 IDs	See Part 3	[1]

CWE and CAPEC

CWE IDs are the KEV catalog's `cwes` field for each CVE (Part 4). CAPEC IDs come from the library's `data/weaknesses/cwe.jsonl` for weakness-to-pattern links and from the `capec` field of `data/attack/technique_profiles.jsonl` for technique-to-pattern links (**Confirmed** [1] [33]). Broad classes (CWE-20, CWE-119, CWE-200, CWE-400) are not expanded. See the CWE Reference and CAPEC Reference.

Weakness	CVEs in this record	CAPEC patterns linked in the library data
CWE-22 Path Traversal	CVE-2019-11510, CVE-2019-19781, CVE-2020-5902, CVE-2018-13379	CAPEC-126 Path Traversal; CAPEC-64; CAPEC-76; CAPEC-78; CAPEC-79
CWE-78 OS Command Injection	CVE-2019-11539	CAPEC-88 OS Command Injection; CAPEC-6; CAPEC-15; CAPEC-43; CAPEC-108

Weakness	CVEs in this record	CAPEC patterns linked in the library data
CWE-77 Command Injection	CVE-2024-21887, CVE-2024-3400	CAPEC-248 Command Injection; CAPEC-136; CAPEC-15; CAPEC-183; CAPEC-40; CAPEC-43; CAPEC-75; CAPEC-76
CWE-94 Code Injection	CVE-2023-3519	CAPEC-242 Code Injection; CAPEC-35; CAPEC-77
CWE-287 Improper Authentication	CVE-2020-12812, CVE-2021-34523, CVE-2020-0688	CAPEC-115 Authentication Bypass; CAPEC-114; CAPEC-151; CAPEC-194; CAPEC-22; CAPEC-57; CAPEC-593; CAPEC-633; CAPEC-650; CAPEC-94
CWE-306 Missing Authentication for Critical Function	CVE-2019-5591, CVE-2022-1388	CAPEC-12; CAPEC-166; CAPEC-216; CAPEC-36 Using Unpublished Interfaces or Functionality; CAPEC-62
CWE-434 Unrestricted Upload of Dangerous File Type	CVE-2021-31207	CAPEC-1 Accessing Functionality Not Properly Constrained by ACLs
CWE-502 Deserialization of Untrusted Data	CVE-2021-44228	CAPEC-586 Object Injection
CWE-917 Expression Language Injection	CVE-2021-45046, CVE-2022-26134	None in the library data
CWE-918 Server-Side Request Forgery	CVE-2021-34473	CAPEC-664 Server Side Request Forgery
CWE-1188 Insecure Default Initialization	CVE-2023-6448	The library's only linked pattern (CAPEC-665, Thunderbolt protection flaws) does not fit a PLC default password; see technique `T1078.001` below

Technique (Iranian use)	CAPEC patterns linked in the library data
`T1110.003` Password Spraying (AA24-290A, Peach Sandstorm)	CAPEC-565 Password Spraying
`T1110` Brute Force (AA23-335A, AA24-290A)	CAPEC-112 Brute Force
`T1078.001` Default Accounts (Unitronics PLCs, AA22-320A)	CAPEC-70 Try Common or Default Usernames and Passwords
`T1133` External Remote Services (AA24-241A, AA24-290A)	CAPEC-555 Remote Services with Stolen Credentials
`T1566.001` and `T1566.002` Spearphishing (seven groups each, AA22-055A)	CAPEC-163 Spear Phishing
`T1505.003` Web Shell (seven groups, AA20-259A, AA24-241A)	CAPEC-650 Upload a Web Shell to a Web Server
`T1036.005` Match Legitimate Resource Name or Location (eight groups)	CAPEC-543 Counterfeit Websites; CAPEC-616 Establish Rogue Location

D3FEND countermeasures

The advisory guidance in Part 8 maps to these D3FEND countermeasures, each with a library page; the "library maps it to" column is the library's own `d3fend_countermeasures.jsonl` and `technique_to_d3fend.jsonl` (D3FEND 1.6.0). The live D3FEND pages for Multi-factor Authentication, Change Default Password and Software Update were opened; the live slug for the first is `d3f:Multi-factorAuthentication` (**Confirmed** [85] [1]). See the D3FEND Reference.

Advisory guidance	D3FEND countermeasure	Library maps it to (Iran-relevant techniques)	Sources
Phishing-resistant MFA	Multi-factor Authentication (`D3-MFA`)	`T1110.003`; not `T1621` or `T1133`	[18] [21] [25] [27]
Replace default passwords	Change Default Password (`D3-CDP`)	`T1078.001`, `T1110.003`	[23] [27]
NIST-aligned password policy	Strong Password Policy (`D3-SPP`)	`T1110.003`	[25]
Patch KEV and internet-facing systems	Software Update (`D3-SU`), Asset Vulnerability Enumeration (`D3-AVE`), Software Inventory (`D3-SWI`)	not `T1190` (a gap)	[18] [19] [21] [24] [27]
Reduce exposure, filter and allowlist remote access	Inbound Traffic Filtering (`D3-ITF`), Network Traffic Filtering (`D3-NTF`), Disable Remote Access (`D3-DRA`)	`D3-ITF` maps `T1190`	[21] [23] [27]
Application control	Executable Allowlisting (`D3-EAL`)	`T1204.002`, `T1505.003`	[18]
Impossible travel and unusual logons	User Geolocation Logon Pattern Analysis (`D3-UGLPA`)	`T1021.001`, `T1190`, `T1572`	[25]
Monitor new accounts	Domain Account Monitoring (`D3-DAM`), Local Account Monitoring (`D3-LAM`)	`T1136.002`, `T1078.002`; `T1136.001`	[21]
Monitor new scheduled tasks	Scheduled Job Analysis (`D3-SJA`)	not `T1053.005` (a gap)	[21]
RDP monitoring	Remote Terminal Session Detection (`D3-RTSD`)	`T1021.001`	[21]
Block tunneling egress (ngrok and similar)	Outbound Traffic Filtering (`D3-OTF`)	`T1572`, `T1105`, `T1071.001`	[21] [24]
External-sender banners, link hygiene	Email Filtering (`D3-EF`), URL Reputation Analysis (`D3-URA`)	`T1566.001`, `T1566.002`, `T1204.001`	[18]
Disable departing accounts	Account Locking (`D3-AL`)	the `T1078` family	[25]
Back up data and PLC logic	Restore Configuration (`D3-RC`), Restore File (`D3-RF`)	`D3-RF` maps `T1486`	[19] [23] [27]
No countermeasure in the library data	none	`T1059.001` PowerShell	[1]

Navigator layers

The study's techniques are collected in two ATT&CK Navigator layers, listed with the library's other curated layers on the Navigator layers page:

- Enterprise: `navigator/library-layers/threat-actors-iran.json`, 260 techniques in 313 technique-tactic entries. Open it in the ATT&CK Navigator. The score is the number of Iran-attributed groups plus the number of joint advisories that name the technique (maximum 18): a count of public reporting, not prevalence or severity. Each comment names the groups and the study's source numbers.
- ICS: `navigator/case-studies/threat-actors-iran-ics.json`, domain `ics-attack`, 24 techniques in 28 technique-tactic entries, scored by the number of the three groups (CyberAv3ngers, OilRig, APT33) linked to each. Open it in the ATT&CK Navigator. Curated layers under `navigator/library-layers/` must be `enterprise-attack`, so the ICS layer sits with the other ICS case-study layers.

Notes on the library's data

- **Count drift against live ATT&CK.** The live VOID MANTICORE page lists 75 technique rows, 25 of them citing the HomeLand Justice campaign, and campaign-derived software (CHIMNEYSWEEP, Impacket, ftp among them); the library data holds 63 direct techniques and no group-level software. The live page appears to fold in campaign relationships the library's group counts leave out (**Confirmed** for the counts [6] [1]; the cause is **Inferred**). The live CyberAv3ngers page (version 1.1, modified 7 October 2026) lists 19 ICS technique rows and links three software entries in its description (Studio Logix 5000 Designer, Totally Integrated Automation (TIA) Portal and EcoStruxure Control Expert), whose own ATT&CK software pages returned 404 when this study re-checked them on 11 November 2026, so their IDs are left out here; the library holds 17 techniques and no software for the group, which points to data drift after the library's ATT&CK pull (**Inferred** [12] [1]).
- **The Threat Actors page needs a correction.** Its Iran-Nexus table lists CyberAv3ngers as a "Volt Typhoon lookalike" with ATT&CK ID "None" (ATT&CK has `G1027`), heads OilRig "(Ministry of Intelligence)" and APT33 "(IRGC)" where ATT&CK v20.0 names no service for either, and covers 4 of the 21 Iran-attributed ATT&CK groups (**Confirmed** [14] [1]). This study is the fuller record until that page is updated.

Cross-cutting themes

- **Attribution has layers, and each source picks one.** The same cluster can be "Iranian" in ATT&CK, "IRGC-affiliated" in a joint advisory, "operating for the IRGC Intelligence Organization, moderate confidence" at Mandiant, and named by front company in a Treasury release. A defender's report should carry the layer and the confidence, not just the name. See the Threat Intelligence Reference.
- **Contractors blur state and crime.** Pioneer Kitten serves the Government of Iran but works with ransomware affiliates on the side, which the FBI judges is likely not sanctioned; Najee and Afkar ran their own BitLocker extortion; SamSam's operators were Iranian criminals with no state tie alleged (**Confirmed** [24] [19] [45]). Treat "Iranian" and "state" as two separate claims.
- **Personas are a delivery mechanism.** HomeLand Justice, Karma, Handala, CyberAv3ngers, Holy Souls and Br0k3r are names chosen for an audience. Governments and vendors have tied each to a state organization, but the persona's own claims (device counts, victim lists) are claims (**Confirmed** [20] [75] [76] [23] [70] [24]; the reading is **Inferred**).
- **Legal action names people and fronts but does not stop the work.** Indictments in 2016, 2018, 2019, 2022, 2024 and 2026 and sanctions in 2018, 2020, 2021, 2022, 2024 and 2026 sit beside an activity

record that continues through 2026 (**Inferred** from the timeline). The 2026 superseding Mabna indictment covers activity from 2013 (**Reported** [87] [88]).

- **Old flaws keep working.** Nineteen of the twenty advisory CVEs are now in KEV, most of them added in the catalog's first release on 3 November 2021, and controllers on default passwords need no CVE at all. The record argues for exposure management over novelty hunting (**Inferred** from [33] [23]).
- **Healthcare is in the blast radius more than the crosshairs.** The confirmed hospital cases come through shared infrastructure; the targeting that looks deliberate is research espionage and, in 2026, a supplier (**Inferred** from Part 6). See the Healthcare Security Reference.

Hype and evidence

Some threats in this record read larger than the evidence supports, and some read smaller.

- **Overstated: persona device counts.** Handala claimed it wiped more than 200,000 Stryker devices; TechCrunch reports tens of thousands, and Stryker's own filing gives no count (**Reported** [93] [91]; the filing **Confirmed** [59]). A persona writes for an audience, so its count stays a claim until the victim or a government confirms it (**Inferred**).
- **Overstated: the hospital cases as a hospital campaign.** The confirmed US hospital cases came through the same edge devices, weak passwords and exposed controllers the actors use everywhere, and the authors of AA21-321A assess the actors exploit known vulnerabilities rather than choosing sectors (**Confirmed** [17]). Reading those cases as a campaign aimed at hospitals overstates the record (**Inferred** from Part 6).
- **Understated: IRGC-CEC PLC tampering.** Treasury describes the 2023 Unitronics activity as defacements, but the AA23-335A update records custom ladder logic and disabled upload and download functions, and AA26-097A records manipulated project files and HMI displays and, in one case, disabled shutdown and alarm logic (**Confirmed** [40] [23] [29]). That is interference with process control, not only a defaced screen, and water, building and hospital facility teams should plan for it (**Inferred**).
- **Understated: device-management abuse.** The Stryker reporting points to a legitimate remote-wipe function in a management console, and the company's filing found no ransomware or malware (**Reported** [93]; **Confirmed** [59]; the link between them is **Inferred**). A program that hunts only for malware would miss a fleet wiped through its own console (**Inferred**).

Sources

Every source below was opened on 3 November 2026. Grades apply to the source type: Confirmed for a primary (the publisher of the fact), Reported for secondary coverage or a filing whose text could not be read.

#	Source	URL	Opened	Grade
1	TeamStarWolf library data at commit e85b559 (ATT&CK v20.0): data/attack/groups.jsonl, group_profiles.jsonl, group_to_technique.jsonl, campaign_profiles.jsonl, software_profiles.jsonl, technique_profiles.jsonl, superseded_by.jsonl, ics/, mobile/, d3fend_countermeasures.jsonl, technique_to_d3fend.jsonl; data/weaknesses/; generated mitre/ pages	https://github.com/TeamStarWolf/TeamStarWolf/tree/e85b5594301fe061043fe92040135252c97b695d/data/attack	2026-11-03	Confirmed
2	MITRE ATT&CK, Group G0059 Magic Hound (version 6.1, modified 31 July 2026)	https://attack.mitre.org/groups/G0059/	2026-11-03	Confirmed
3	MITRE ATT&CK, Group G0049 OilRig (version 5.0, modified 31 July 2026)	https://attack.mitre.org/groups/G0049/	2026-11-03	Confirmed
4	MITRE ATT&CK, Group G0069 MuddyWater (version 7.0, modified 31 July 2026)	https://attack.mitre.org/groups/G0069/	2026-11-03	Confirmed
5	MITRE ATT&CK, Group G1059 NIMBUS MANTICORE (version 1.0, created 1 October 2026)	https://attack.mitre.org/groups/G1059/	2026-11-03	Confirmed
6	MITRE ATT&CK, Group G1055 VOID MANTICORE (version 1.0, created 20 April 2026, modified 31 July 2026)	https://attack.mitre.org/groups/G1055/	2026-11-03	Confirmed
7	MITRE ATT&CK, Group G0087 APT39 (version 3.2)	https://attack.mitre.org/groups/G0087/	2026-11-03	Confirmed
8	MITRE ATT&CK, Group G0117 Fox Kitten (version 2.0)	https://attack.mitre.org/groups/G0117/	2026-11-03	Confirmed
9	MITRE ATT&CK, Group G1044 APT42 (version 1.1, modified 22 September 2026)	https://attack.mitre.org/groups/G1044/	2026-11-03	Confirmed
10	MITRE ATT&CK, Group G0064 APT33 (version 2.0)	https://attack.mitre.org/groups/G0064/	2026-11-03	Confirmed
11	MITRE ATT&CK, Group G1030 Agrius (version 1.0)	https://attack.mitre.org/groups/G1030/	2026-11-03	Confirmed
12	MITRE ATT&CK, Group G1027 CyberAv3ngers (version 1.1, modified 7 October 2026)	https://attack.mitre.org/groups/G1027/	2026-11-03	Confirmed
13	MITRE ATT&CK, Campaign C0038 HomeLand Justice (version 1.1)	https://attack.mitre.org/campaigns/C0038/	2026-11-03	Confirmed
14	TeamStarWolf library, THREAT_ACTORS.md at commit e85b559	https://github.com/TeamStarWolf/TeamStarWolf/blob/e85b5594301fe061043fe92040135252c97b695d/THREAT_ACTORS.md	2026-11-03	Confirmed
15	CISA with FBI, AA20-259A, Iran-based actor exploiting VPN and edge vulnerabilities (2020-09-15)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-259a	2026-11-03	Confirmed
16	CISA, AA20-304A, Iranian APT targeting US voter information (2020-10-30)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-304a	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
17	FBI, CISA, ACSC and NCSC, AA21-321A, Iranian Government-Sponsored APT Cyber Actors Exploiting Microsoft Exchange and Fortinet Vulnerabilities (2021-11-17)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-321a	2026-11-03	Confirmed
18	FBI, CISA, CNMF and NCSC, AA22-055A, MuddyWater (2022-02-24)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-055a	2026-11-03	Confirmed
19	FBI, CISA, NSA, CNMF, Treasury, ACSC, CCCS and NCSC, AA22-257A, IRGC-affiliated cyber actors (2022-09-14)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-257a	2026-11-03	Confirmed
20	FBI and CISA, AA22-264A, Iranian state actors' attack on the Government of Albania (2022-09-21)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-264a	2026-11-03	Confirmed
21	CISA and FBI, AA22-320A, Iranian government-sponsored APT compromises a federal network (2022-11-16)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-320a	2026-11-03	Confirmed
22	CISA alert, Exploitation of Unitronics PLCs used in Water and Wastewater Systems (2023-11-28)	https://www.cisa.gov/news-events/alerts/2023/11/28/exploitation-unitronics-plcs-used-water-and-wastewater-systems	2026-11-03	Confirmed
23	FBI, CISA, NSA, EPA, INCD, CCCS and NCSC, AA23-335A, IRGC-affiliated actors exploit PLCs (2023-12-01, updated 2024-12-18)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-335a	2026-11-03	Confirmed
24	FBI, CISA and DC3, AA24-241A, Iran-based actors enabling ransomware (2024-08-28)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-241a	2026-11-03	Confirmed
25	FBI, CISA, NSA, CSE, AFP and ASD's ACSC, AA24-290A, Iranian brute force and credential access (2024-10-16)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-290a	2026-11-03	Confirmed
26	CISA, FBI, DC3 and NSA, Iranian Cyber Actors May Target Vulnerable US Networks and Entities of Interest, web page (2025-06-30)	https://www.cisa.gov/resources-tools/resources/iranian-cyber-actors-may-target-vulnerable-us-networks-and-entities-interest	2026-11-03	Confirmed
27	The same joint fact sheet, PDF	https://www.cisa.gov/sites/default/files/2025-06/joint-fact-sheet-Iranian-cyber-actors-may-target-vulnerable-US-networks-and-entities-of-interest-508c-1.pdf	2026-11-03	Confirmed
28	The same fact sheet as posted by IC3	https://www.ic3.gov/CSA/2025/250630.pdf	2026-11-03	Confirmed
29	FBI, CISA, NSA, EPA, DOE, CNMF and Treasury, AA26-097A, Iranian-affiliated APT disrupts PLCs (2026-04-07, revised 2026-07-22)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a	2026-11-03	Confirmed
30	FBI, Treasury and INCD, Emennet Pasargad advisory (2024-10-30)	https://www.ic3.gov/CSA/2024/241030.pdf	2026-11-03	Confirmed
31	ODNI, FBI and CISA, joint statement on Iranian election influence (2024-08-19)	https://content.govdelivery.com/accounts/USDHSCISA/bulletins/3afd156	2026-11-03	Confirmed
32	CISA, AA18-337A, SamSam ransomware (2018-12-03)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa18-337a	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
33	CISA, Known Exploited Vulnerabilities catalog, JSON feed (catalog version 2026.11.03)	https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-03	Confirmed
34	US Treasury, sm0332, Mabna Institute designation (2018-03-23)	https://home.treasury.gov/news/press-releases/sm0332	2026-11-03	Confirmed
35	US Treasury, sm556, SamSam-related designations (2018-11-28)	https://home.treasury.gov/news/press-releases/sm556	2026-11-03	Confirmed
36	US Treasury, sm1127, APT39 and Rana Intelligence Computing (2020-09-17)	https://home.treasury.gov/news/press-releases/sm1127	2026-11-03	Confirmed
37	US Treasury, jy0494, Emennet Pasargad (2021-11-18)	https://home.treasury.gov/news/press-releases/jy0494	2026-11-03	Confirmed
38	US Treasury, jy0941, MOIS and Minister Khatib (2022-09-09)	https://home.treasury.gov/news/press-releases/jy0941	2026-11-03	Confirmed
39	US Treasury, jy0948, Najee Technology and Afkar System Yazd (2022-09-14)	https://home.treasury.gov/news/press-releases/jy0948	2026-11-03	Confirmed
40	US Treasury, jy2072, IRGC Cyber-Electronic Command officials (2024-02-02)	https://home.treasury.gov/news/press-releases/jy2072	2026-11-03	Confirmed
41	US Treasury, jy2292, IRGC-CEC front companies (2024-04-23)	https://home.treasury.gov/news/press-releases/jy2292	2026-11-03	Confirmed
42	US Treasury, jy2621, election interference designations (2024-09-27)	https://home.treasury.gov/news/press-releases/jy2621	2026-11-03	Confirmed
43	FBI, Iranian DDoS Attacks (wanted notice)	https://www.fbi.gov/wanted/cyber/iranian-ddos-attacks	2026-11-03	Confirmed
44	FBI, Iranian Mabna Hackers (wanted notice)	https://www.fbi.gov/wanted/cyber/iranian-mabna-hackers	2026-11-03	Confirmed
45	FBI, SamSam Subjects (wanted notice)	https://www.fbi.gov/wanted/cyber/samsam-subjects	2026-11-03	Confirmed
46	FBI, Monica Elfriede Witt (wanted notice)	https://www.fbi.gov/wanted/counterintelligence/monica-elfriede-witt	2026-11-03	Confirmed
47	FBI, Iranian Cyber Actors (wanted notice, Ahmadi and others)	https://www.fbi.gov/wanted/cyber/iranian-cyber-actors	2026-11-03	Confirmed
48	FBI, Masoud Jalili (wanted notice)	https://www.fbi.gov/wanted/cyber/masoud-jalili	2026-11-03	Confirmed
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The terms for every MITRE dataset used here are recorded in THIRD_PARTY_NOTICES.md.

Related references

- Threat Actors (see the correction note in Part 9)
- Malware Families
- Notable Incidents
- CVE Reference
- Healthcare Security Reference, ICS and OT Security Reference, ICS ATT&CK Atlas
- Ransomware Defense Reference, Identity Security Reference
- Threat Intelligence Reference, Threat Hunting Reference, Vulnerability Prioritization Reference
- CWE Reference, CAPEC Reference, D3FEND Reference
- Generated MITRE pages: groups Magic Hound G0059, OilRig G0049, MuddyWater G0069, NIMBUS MANTICORE G1059, VOID MANTICORE G1055, APT39 G0087, Fox Kitten G0117, APT42 G1044, APT33 G0064, Agrius G1030 and Silent Librarian G0122; campaigns C0010, C0038 HomeLand Justice, C0042 Outer Space and C0044 Juicy Mix; software Shamoon S0140, ZeroCleare S1151, Pay2Key S0556, POWERSTATS S0223 and IOControl S9046; techniques Exploit Public-Facing Application T1190, External Remote Services T1133, Password Spraying T1110.003, Multi-Factor Authentication Request Generation T1621, Default Accounts T1078.001 and Web Shell T1505.003
- Related case studies: All case studies; China-Attributed Threat Actors, Russia-Attributed Threat Actors, Other State-Linked Threat Actors, United States-Attributed Threat Actors, ICS Attacks and Advisories,

2024 to 2026, Log4Shell, Operation Exchange Marauder, Stuxnet, Change Healthcare 2024, The Year 2019 in Intrusions, The Year 2020 in Intrusions, APT41: The Public Record, Sandworm Operations

- Navigator layers